

The birth–death–catastrophe process: further results, and a burst-aware model of viral replication

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1 Introduction

The standard models of within-host infection treat the infected cell as a black box. In the Basic Model of Viral Replication, a cell that has been infected releases new particles at a constant rate p until it is removed at a constant rate d_I ; nothing about the cell's interior enters the dynamics, and the two constants are phenomenological — parameters to be fitted, not quantities to be derived. For a budding virus this is at least a defensible caricature. For a lytic pathogen it is plainly wrong in mechanism, whatever it may achieve by fitting: such a pathogen grows inside the cell, and releases nothing until the cell ruptures, at which point it releases everything at once. *Yersinia pestis* in the macrophage, *Francisella tularensis* in the macrophage, the lytic phage in its bacterium — all follow the intracellular replication-and-lysis picture, and all are poorly served by a constant drip.

This chapter asks what the constant-rate assumption should be replaced with, and what the replacement implies. The intracellular model was set up in Chapter 3: the birth–death–catastrophe process, in which the pathogen count within one cell is born at rate β per particle, dies at rate μ per particle, and triggers the rupture of the cell at rate δ per particle. Chapter 3 derived the principal single-cell quantities — the no-rupture probability $I(t)$, the mean and second moment $J(t)$, $K(t)$, and the cumulative mean release $V(t)$. What it did not derive is the full distribution of the process, the statistics of the rupture event itself, or the passage from one cell to a population of them. These are the subjects of the present chapter.

Four questions organise it.

1. What is the complete single-cell description? The state probabilities at every time, the quasi-stationary distribution of the load, and the laws of the burst time and the burst size (§§5–8).
2. How do several founders in one cell, and chains of cells, behave? Multiplicity of infection (§9) and chained immediate transfer (§10).
3. How should the single-cell theory be embedded in a population model? Why the constant-rate substitution fails (§11), and the age-structured renewal model that replaces it (§12).
4. What does the replacement imply? Effective parameters and the invariance of R_0 , and what population data can and cannot identify (§13); the flooding advantage of bursting and its trade-off against invasion speed (§14); the spectrum of release models between bursting and budding (§15); and the boundary of the lytic claims, drawn against HIV (§16).

The answers, in advance and in brief, are these. The single-cell process is solvable far beyond its moments: the state probabilities are geometric at every time, and their conditional limit — the quasi-stationary distribution — is geometric with ratio $1/a$. The burst-size distribution, conditioned on bursting, is *exactly the same law*: a cell's rupture size is distributed as the load it would have had, had it kept growing forever. A population of such cells is not a constant-rate system but an age-structured one, governed by two closed-form kernels — the survival kernel $\hat{I}(a)$ and the release kernel $g(a) = \delta K(a)$ — and the renewal model built on them contains classical BMVR as an exact exponential-phase projection. Under that projection, the effective release rate p_{eff} depends on the epidemic growth rate, falling by orders of magnitude from the stationary phase to the acute phase; R_0 itself is invariant to bursting at fixed total output; and the three intracellular rates cannot be recovered from the two BMVR parameters alone. At fixed output, bursting lowers the probability of early extinction — relative to matched budding — if and only if $1/\delta < 1/\beta + 1/\mu$, at the cost of a slower deterministic invasion. And the honest boundary of all this is that HIV, which buds, is not

a BDC pathogen; the chapter ends by saying precisely what the lytic theory does and does not offer it.

1.1 Two varieties of rupture: killing and catastrophe

Before we begin, it is important to state a distinction between two varieties of the model. When a rupture event occurs, the process goes to a state. In the literature, this state is sometimes chosen to be 0, and in other cases there is a separate rupture state defined, sometimes labelled R . The analysis does differ between the two options. And so for clarity, in this chapter we will refer to the 0-state case as “killing”; the distinct rupture state as “catastrophe”. The distinction matters most in §§3–5, where both conventions are carried in parallel; from §6 onward the catastrophe convention is primary.

2 The single-cell BDC process: recap and reference quantities

Chapter 3 defined the birth–death–catastrophe process and derived its principal quantities. This chapter builds on that foundation, and several of the results it needs are restated here — in corrected and simplified form — so that what follows is self-contained. Section 2.1 fixes the notation used throughout; the remaining sections collect the closed forms that will be quoted repeatedly. Derivations are sketched; the full accounts live in Chapter 3 and, for the corrected items, in the remarks.

2.1 Canonical notation

The notation of this chapter is collected in Table 1. It resolves the collisions that arise the moment the intracellular and population models are joined: β vs λ for the birth rate; δ for the catastrophe rate vs the infected-cell removal rate; I and V for probabilities and expectations vs cell and virion counts; H vs R for the rupture state.

Symbol	Meaning	Notes
β, μ, δ	birth, death, catastrophe rates (per capita)	λ is not used for birth
$\mathbf{X}_t$	intracellular count (CTMC)	
R	absorbing rupture state	
$\mathbf{W}_t$	released count from one cell	0 until burst
τ	burst time (possibly ∞)	
$\mathcal{K}$	burst size (random variable)	never bare K , since $K(t) = \mathbb{E}(\mathbf{X}_t^2)$
$I(t)$	$\mathbb{P}\{\text{no burst by } t\}$	not an infected-cell count
$D(t)$	$\mathbb{P}\{\text{cleared internally, no burst}\}$	$= p_0(t)$
$\widehat{I}(t)$	$\mathbb{P}\{\mathbf{X}_t \notin \{0, R\}\} = I(t) - D(t)$	productive infection
$J(t), K(t)$	$\mathbb{E}(\mathbf{X}_t), \mathbb{E}(\mathbf{X}_t^2)$	
$V(t)$	$\mathbb{E}(\mathbf{W}_t)$, cumulative mean release	not a free-virion count
a, b	characteristic roots, $b < 1 < a$	
A, B, θ	$a - 1, 1 - b, \beta(a - b)$	
L	$a(1 - b) = a - \mu/\beta$	flooding parameter
T	target-cell count (held fixed)	population model
γ	infection rate	not β
$\mathcal{I}, \mathcal{V}$	infected cells, free virions	population counts
c	virion clearance	
$d_{\mathcal{I}}$	infected-cell removal (phenomenological)	not δ
$p, p_{\text{eff}}(r)$	constant / effective release rate	
$g(a)$	release kernel (expected flux at age a)	
q	$\gamma T / (\gamma T + c)$, infection success	branching

Table 1: Canonical notation for this chapter.

2.2 Process definition and the joint process

One cell, one intracellular population. The count $\mathbf{X}_t$ is a continuous-time Markov chain on $\{0, 1, 2, \dots\} \cup \{R\}$ which, from any state $n \geq 1$, jumps

$$\begin{aligned}
n &\rightarrow n + 1 && \text{at rate } \beta n \quad (\text{birth}), \\
n &\rightarrow n - 1 && \text{at rate } \mu n \quad (\text{death}), \\
n &\rightarrow R && \text{at rate } \delta n \quad (\text{catastrophe}).
\end{aligned}$$

Both 0 and R are absorbing: the population either clears itself internally, or the cell ruptures. Unless stated otherwise we start from a single founder, $X_0 = 1$. The process of classical references [1, 2, 3] is slightly more general (immigration, distributed catastrophe sizes); the specialisation above is the one with the closed-form family we need.

The rupture event is what matters for applications, so we record the release count alongside the load. Let τ be the time of catastrophe (possibly $\tau = \infty$ if the population clears internally first). The released count $\mathbf{W}_t$ is

$$\mathbf{W}_t = \begin{cases} 0, & t < \tau, \\ X_{\tau-}, & t \geq \tau, \end{cases}$$

that is, at rupture the whole pre-burst load is dumped and then held. All single-cell release quantities in this chapter are functionals of the joint process $(\mathbf{X}_t, \mathbf{W}_t)$.

2.3 Roots and the workhorse identity

Write

$$\eta = \frac{\beta + \mu + \delta}{2\beta}, \quad a = \eta + \sqrt{\eta^2 - \frac{\mu}{\beta}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\beta}}, \quad (1)$$

the two roots of the quadratic $\beta f^2 - (\beta + \mu + \delta)f + \mu$ that will appear in the next subsection, and set

$$A := a - 1 > 0, \quad B := 1 - b > 0, \quad \theta := \beta(a - b) = \sqrt{(\beta + \mu + \delta)^2 - 4\beta\mu}. \quad (2)$$

Vieta's relations give $ab = \mu/\beta$ and $a + b = (\beta + \mu + \delta)/\beta$, and the identity that will do most of the algebraic work in this chapter is

$$AB = (a - 1)(1 - b) = (a + b) - ab - 1 = \frac{\delta}{\beta}, \quad A + B = a - b. \quad (3)$$

That $b < 1 < a$ always is immediate from $\beta(1 - a)(1 - b) = \beta(1 - (a + b) + ab) = -\delta < 0$, which places 1 strictly between the two roots.

2.4 Three fixation functions

There are three absorbing-behaviour probabilities, not two, and it pays to name all of them:

Definition 2.1.

$$I(t) = \mathbb{P}\{\text{no burst by } t\}, \quad D(t) = \mathbb{P}\{\text{internal extinction by } t, \text{ no burst}\}, \quad (4)$$

$$\hat{I}(t) = \mathbb{P}\{\mathbf{X}_t \notin \{0, R\}\} = I(t) - D(t), \quad (5)$$

the last being the probability that the cell is still productively infected.

Both I and D satisfy the same Riccati equation. For I , condition on the first event of the founder: a birth (rate β) replaces the one lineage by two independent ones, both of which must avoid bursting, contributing I^2 ; a death (rate μ) makes bursting impossible, contributing 1; a catastrophe (rate δ) contributes 0. Hence

$$\frac{dI}{dt} = \beta(I^2 - I) + \mu(1 - I) + \delta(0 - I) = \beta(I - a)(I - b) = \mu - (\beta + \mu + \delta)I + \beta I^2, \quad (6)$$

with $I(0) = 1$. The same first-step decomposition with “cleared internally, no burst” in place of “no burst” gives the identical equation for D , now with $D(0) = 0$. Solving (6) by separation and writing $w := e^{\theta t} \geq 1$,

$$I(t) = \frac{aB + bAw}{B + Aw}, \quad D(t) = \frac{ab(w - 1)}{aw - b}, \quad \hat{I}(t) = \frac{(a - b)^2 w}{(B + Aw)(aw - b)}. \quad (7)$$

The limits are $I(\infty) = D(\infty) = b$, $\hat{I}(\infty) = 0$, with $\hat{I}(0) = 1$ and $\hat{I}'(0) = -(\mu + \delta)$ — at the instant of founding, the productive state is lost only by death or catastrophe of the single founder.

Remark 2.2 (Correction of the Chapter 3 formula for $\hat{I}$). Chapter 3 gives $\hat{I}(t) = \hat{\eta} / (1 - (1 - \hat{\eta})e^{\hat{\eta}\beta t})$ with $\hat{\eta} = (\beta + \mu + \delta)/\beta$, derived from the ODE $\hat{I}' = -(\beta + \mu + \delta)\hat{I} + \beta\hat{I}^2$. The branching term is wrong: on a birth, the two daughter lineages must avoid bursting *and* not both go extinct, an event of probability $I^2 - D^2$, not $\hat{I}^2$. The error is invisible when $\mu = 0$ (then $D \equiv 0$ and $\hat{I} = I$), which is presumably why it survived. Numerically, at $(\beta, \mu, \delta) = (1, 0.2, 0.05)$ and $t = 1$: the correct value is $\hat{I} \approx 0.8087$; the thesis formula gives ≈ 0.6675 . An independent confirmation arrives in §5: the state probabilities $p_n(t)$ sum over $n \geq 1$ to exactly the right-hand side of (7).

2.5 Moments and mean release

The hazard of bursting at time t is δ times the current load, so $I'(t) = -\delta J(t)$; inserting the Riccati equation (6) then gives the mean

$$J(t) = \mathbb{E}(\mathbf{X}_t) = -\frac{1}{\delta} \frac{dI}{dt} = \frac{(a-b)^2 w}{(B+Aw)^2} = -\frac{\beta}{\delta} (I-a)(I-b), \quad (8)$$

with $J(0) = 1$ now manifest. The second moment,

$$K(t) = \mathbb{E}(\mathbf{X}_t^2) = \left[1 + \frac{2\beta}{\delta}(1-I)\right]J = \frac{2\beta}{\delta}(\kappa - I)J, \quad \kappa = 1 + \frac{\delta}{2\beta}, \quad (9)$$

follows from the moment hierarchy: one checks by differentiation, using (6), that the right-hand side satisfies $K' = 2(\beta - \mu)K + (\beta + \mu)J - \delta\mathbb{E}(\mathbf{X}_t^3)$ with the third moment eliminated exactly as in Chapter 3.

The cumulative mean release $V(t) = \mathbb{E}(\mathbf{W}_t)$ increases only at rupture events, and a rupture from state n occurs at rate δn and releases n particles, so the expected release flux is $\delta\mathbb{E}(\mathbf{X}_t^2)$:

$$V'(t) = \delta K(t), \quad V(0) = 0. \quad (10)$$

Integrating with (8)–(9) yields the simplification

$$V(t) = (1-I)\left[1 + \frac{\beta}{\delta}(1-I)\right] = \frac{\beta}{\delta} \left((I-\kappa)^2 - (1-\kappa)^2\right), \quad (11)$$

a quadratic in $(1-I)$ alone — no J , no separate μ . Chapter 3's form $V = \frac{\beta-\mu}{\delta}(1-I) - J + 1$ is the same quantity, less happily dressed; the prefactor that clings to J in Chapter 3 cancels identically by (3).

The lifetime mean release and the conditional mean burst size are

$$V_\infty = \frac{a(1-b)}{a-1} = \frac{aB}{A} = \frac{\beta-\mu}{\delta}(1-b) + 1, \quad \mathbb{E}(\mathcal{K} \mid \text{burst}) = \frac{V_\infty}{1-b} = \frac{a}{a-1}. \quad (12)$$

A small point with large consequences: V_∞ is $\mathbb{E}(\mathcal{K}\mathbf{1}\{\text{burst}\})$, and so *includes* the realisations that cleared internally and released nothing (a mass b at 0). The conditional mean removes them. When $\mu = 0$: $b = 0$, $a = 1 + \delta/\beta$, and $V_\infty = 1 + \beta/\delta$ coincides with the conditional mean — a coincidence that, as we shall see, one should not lean on.

2.6 Second moment of the release

For the population-level variance in §17.1 we will need $\mathbb{E}(\mathbf{W}_t^2)$. The release variable jumps once, from 0 to $X_{\tau-}$, so $\frac{d}{dt}\mathbb{E}(\mathbf{W}_t^2) = \delta\mathbb{E}(\mathbf{X}_t^3)$. The third moment is eliminated via the second-moment

ODE, $K' = 2(\beta - \mu)K + (\beta + \mu)J - \delta\mathbb{E}(\mathbf{X}_t^3)$, giving

$$\frac{d}{dt}\mathbb{E}(\mathbf{W}_t^2) = 2(\beta - \mu)K + (\beta + \mu)J - K' = \frac{d}{dt}\left(\frac{2(\beta - \mu)}{\delta}V - \frac{\beta + \mu}{\delta}I - K\right),$$

where the second equality uses $V' = \delta K$ and $J = -\delta^{-1}I'$ — note the sign. Integrating from 0 with $(I, J, K, V, \mathbb{E}(W^2))(0) = (1, 1, \mathbf{1}, 0, 0)$ — in particular $K(0) = \mathbb{E}(X_0^2) = 1$, not 0 —

$$\mathbb{E}(\mathbf{W}_t^2) = \frac{2(\beta - \mu)}{\delta}V - \frac{\beta + \mu}{\delta}I - K + \frac{\beta + \mu}{\delta} + 1, \quad (13)$$

and $\text{Var}(\mathbf{W}_t) = \mathbb{E}(\mathbf{W}_t^2) - V^2$.

Remark 2.3 (Correction and check). Chapter 3 §8 has wrong signs on I and K and omits the constant $+1$; both errors trace to the two slips above (the sign of $J = -\delta^{-1}I'$ and the value of $K(0)$). An independent verification: take $\mu = 0$ and $t \rightarrow \infty$. Then $I, K \rightarrow 0$ and $V \rightarrow 1 + \beta/\delta$, so (13) gives $\mathbb{E}(W_\infty^2) = 2\beta^2/\delta^2 + 3\beta/\delta + 1$. But when $\mu = 0$ the burst size is geometric on $\{1, 2, \dots\}$ with success probability $s = \delta/(\beta + \delta)$ (proved in §7.2), whose second moment is $(2 - s)/s^2 = 2\beta^2/\delta^2 + 3\beta/\delta + 1$. Exact agreement.

3 Probability Generating Function PDEs

The probability generating function (probability generating function) of the intracellular count is defined by

$$G(z, t) = p_0(t) + p_1(t)z + p_2(t)z^2 + p_3(t)z^3 + \dots,$$

where $p_i(t) = \Pr\{\mathbf{X}_t = i\}$. The following PDEs are derived from the forward Kolmogorov equations associated with the two rupture conventions.

Catastrophe

$$\frac{\partial G(z, t)}{\partial t} = (\beta z^2 - (\beta + \mu + \delta)z + \mu) \frac{\partial G}{\partial z} \quad (14)$$

Killing

$$\frac{\partial G(z, t)}{\partial t} = \delta J(t) + (\beta z^2 - (\beta + \mu + \delta)z + \mu) \frac{\partial G}{\partial z} \quad (15)$$

A word of warning before we proceed. In the catastrophe convention, ruptured realisations are absorbed into the separate state R and no longer contribute to the count, so $G(1, t) = I(t) \leq 1$: the probability generating function is *defective*, and its total mass decays as cells burst. In the killing convention the rupture transition lands in state 0 and is counted, so the probability generating function remains proper: $G(1, t) = 1$ for all t . We will keep track of which is which.

Derivation

It will be sufficient to show a derivation of the “killing” equation; the “catastrophe” version is similar and simpler.

Take the forward Kolmogorov equations:

$$\begin{cases} \frac{dp_i}{dt} = \beta(i-1)p_{i-1} + \mu(i+1)p_{i+1} - (\beta + \mu + \delta)ip_i, & i > 0 \\ \frac{dp_0}{dt} = \mu p_1 + \delta \sum_{j=1}^{\infty} j p_j. \end{cases}$$

($p_i = \Pr\{\mathbf{X}_t = i\}$, the probability the process is in state i). In the latter equation, the summation represents all of the possible rupture transitions: from any positive state, a collapse to 0 can take place. This summation happens to express the mean, $\mathbb{E}(\mathbf{X}_t)$, which is elsewhere labelled $J(t)$.

In the catastrophe version, since rupture causes transition to a separate state, this summation term is not present, and the first equation holds for all $i \geq 0$ with $p_0(t) \equiv 0$.

The derivation proceeds by manipulating the equations to produce terms containing the probability generating function, $G(z, t) = \sum_{i=0}^{\infty} p_i z^i$. It begins by multiplying each side of every i th equality by z^i . In the case of $i = 0$, $z^0 = 1$, so there is no change.

$$\begin{cases} \frac{dp_i z^i}{dt} = \beta(i-1)p_{i-1}z^i + \mu(i+1)p_{i+1}z^i - (\beta + \mu + \delta)ip_i z^i, & i > 0 \\ \frac{dp_0}{dt} = \mu p_1 + \delta J(t). \end{cases}$$

Then, each equation is combined in a sum over all values of i for which there is a positive term:

$$\sum_{i=0}^{\infty} \frac{\partial p_i z^i}{\partial t} = \delta J(t) + \sum_{i=0}^{\infty} \left[\beta(i-1)p_{i-1}z^i + \mu(i+1)p_{i+1}z^i - (\beta + \mu + \delta)ip_i z^i \right]$$

Some index-shifting converts this into

$$\frac{\partial G(z, t)}{\partial t} = \delta J(t) + \beta z^2 \sum_{i=1}^{\infty} i p_i z^{i-1} + \mu \sum_{i=1}^{\infty} i p_i z^{i-1} - (\beta + \mu + \delta) z \sum_{i=1}^{\infty} i p_i z^{i-1}$$

and since

$$\sum_{i=1}^{\infty} i p_i z^{i-1} = \frac{\partial}{\partial z} \sum_{i=1}^{\infty} p_i z^i = \frac{\partial G(z, t)}{\partial z},$$

the above becomes

$$\frac{\partial G(z, t)}{\partial t} = \delta J(t) + \beta z^2 \frac{\partial G}{\partial z} + \mu \frac{\partial G}{\partial z} - (\beta + \mu + \delta) z \frac{\partial G}{\partial z},$$

equivalent to (15). Dropping the $\delta J(t)$ term — no rupture flux enters any state of the count in the catastrophe convention — leaves (14).

4 Identifying the PGFs

$G(z, t)$, the probability generating function, is a powerful tool for understanding the behaviour of the system and its probable outcomes. Because it can be written as a polynomial sum in which each coefficient is the probability of the process being in a specific state at a given time, valuable information on the process can be extracted.

As,

$$G(z, t) = p_0(t) + p_1(t)z + p_2(t)z^2 + p_3(t)z^3 + \dots,$$

The mean, $J(t)$ can be found with $\partial_z G(z, t)|_{z=1}$:

$$\mathbb{E}(\mathbf{X}_t) = p_1(t) + 2p_2(t) + 3p_3(t) + \dots;$$

Individual state-probabilities can be extracted in the following way. Differentiate with respect to z , n times,

$$\frac{\partial^n G(z, t)}{\partial z^n} = n! \cdot p_n(t) + (n+1)! \cdot p_{n+1}(t)z + \frac{(n+2)!}{2} \cdot p_{n+2}(t)z^2 + \dots,$$

then evaluate at $z = 0$ and divide by $n!$:

$$\frac{1}{n!} \cdot \frac{\partial^n G(z, t)}{\partial z^n} \Big|_{z=0} = p_n(t). \quad (16)$$

In this way the full dynamic distribution of the system can be elucidated.

Though to do this, we must first obtain the explicit formula $G(z, t)$. There will be some difference between the two cases, but the process is similar — both (14) and (15) can be solved with the method of characteristics.

4.1 Method of characteristics

Generally, first order, linear PDEs of the type we are dealing can be solved as a family of ODEs. The method begins by introducing a change of variables $(z, t) \rightarrow (s, \tau)$;

$$G(z, t) \equiv G(z(s, \tau), t(s, \tau)) = G(s, \tau).$$

It proceeds by seeking solutions on characteristic lines of constant s : $G(\tau)$. An initial curve must be fixed, from which these solutions will progress. The selection is somewhat arbitrary; many options will work, though a good choice can keep the analysis more simple. There are some hard conditions. For example, the initial line must not be parallel in the s, τ space to the progressing solutions. In cases such as ours, it is often appropriate to use

$$z(s, 0) = s \quad t(s, 0) = 0 \quad (17)$$

Catastrophe

Assuming the probability generating function changes with only one variable — $G(\tau(z, t))$ — we can write down the characteristic equations as ODEs. First, consider

$$\frac{dG}{d\tau} = \left\{ \frac{dt}{d\tau} \right\} \frac{\partial G}{\partial t} + \left\{ \frac{dz}{d\tau} \right\} \frac{\partial G}{\partial z}.$$

Comparing with the catastrophe equation, (14), there is a similarity,

$$0 = \{-1\} \frac{\partial G}{\partial t} + \left\{ \beta z^2 - (\beta + \mu + \delta)z + \mu \right\} \frac{\partial G}{\partial z},$$

and we can extract the three characteristic ODEs:

$$\frac{dG}{d\tau} = 0, \quad \frac{dt}{d\tau} = -1, \quad \frac{dz}{d\tau} = \beta z^2 - (\beta + \mu + \delta)z + \mu.$$

The first of these, which will differ in the killing case, tells us that

$$G = \phi(s),$$

ϕ constant on curves of unchanging s .

In both cases, the initial condition on G will be

$$G(z, 0) = z$$

because starting with a single individual means $p_1(0) = 1$ and $p_{i \neq 1}(0) = 0$. Therefore, $G = \phi(s)$ in conjunction with our choice of initial curve, $z(s, 0) = s$, allows $\phi(s)$ to be fixed as s ;

$$G(s, \tau) = s.$$

Moving to the second characteristic ODE:

$$\frac{dt}{d\tau} = -1 \implies t = -\tau + C_1(s) \text{ (constant for unchanging } s).$$

Our initial condition, $t(s, 0) = 0$ fixes $C_1(s) = 0$, so $t = -\tau$.

The remaining characteristic equation with z happens to be the same form as an ODE describing the no-rupture probability, $I(t)$.

$$\frac{dz}{d\tau} = \beta z^2 - (\beta + \mu + \delta)z + \mu \quad (18)$$

What differs is the initial condition. In the case of, $I(t)$, at time 0 it is 1 — rupture having definitely not occurred; with $z(s, t)$, our chosen initial condition is $z(s, 0) = s$. Hence, the solution to (18) is very similar to the formula for $I(t)$, though in places where there is seen a one, here an s is found:

$$z = \frac{a(s - b) - b(s - a)e^{\beta(a-b)\tau}}{(s - b) - (s - a)e^{\beta(a-b)\tau}}$$

Also, in this it is τ rather than t .

Since $G(z, t) = s(z, t)$, we can obtain the final solution in one step: Rearrange the above for s , and place $-t$ at each τ . By a happenstance of symmetry, this turns out looking quite the same:

$$s = \frac{a(z - b) - b(z - a)e^{\beta(a-b)t}}{(z - b) - (z - a)e^{\beta(a-b)t}}. \quad (19)$$

And so we have it, by the somewhat dark art of characteristics, the probability generating function emerges as

$$G(z, t) = \frac{a(z - b) - b(z - a)e^{\beta(a-b)t}}{(z - b) - (z - a)e^{\beta(a-b)t}}.$$

As will be seen, in obtaining the state probabilities via repeated partial differentiation in z , it is advantageous to write the formula as

$$G(z, t) = \frac{ab(1 - e^{\beta(a-b)t}) - z(a - be^{\beta(a-b)t})}{(b - ae^{\beta(a-b)t}) - z(1 - e^{\beta(a-b)t})}$$

For now, let us move on to the modification that is introduced in the killing case.

Killing

When rupture collapses the process to 0, it may have happened at any positive state. As we've seen, because of this, the forward Kolmogorov equation describing $p_0(t)$ includes a separate term related to every natural number. Thankfully, these do compress neatly into an expression of the mean, $\mathbb{E}(\mathbf{X}_t) = J(t)$. Hence, in the PDE, (15), there is an inhomogeneous term: $\delta J(t)$.

Making the same comparison as in the catastrophe case,

$$\begin{aligned}\frac{dG}{d\tau} &= \left\{ \frac{dt}{d\tau} \right\} \frac{\partial G}{\partial t} + \left\{ \frac{dz}{d\tau} \right\} \frac{\partial G}{\partial z} \\ -\delta J(t) &= \{-1\} \frac{\partial G}{\partial t} + \left\{ \beta z^2 - (\beta + \mu + \delta)z + \mu \right\} \frac{\partial G}{\partial z}\end{aligned}$$

we see the characteristic equations are largely the same, just that the first picks up this extra component:

$$\frac{dG}{d\tau} = -\delta J(-\tau), \quad \frac{dt}{d\tau} = -1, \quad \frac{dz}{d\tau} = \beta z^2 - (\beta + \mu + \delta)z + \mu.$$

From previous work, we know

$$-\delta J(t) = \frac{dI(t)}{dt}.$$

Simply because $dt = -d\tau$, it follows that

$$\delta J(-\tau) = \frac{dI(-\tau)}{d\tau}.$$

And so, looking to the first characteristic equation,

$$\frac{dG}{d\tau} = \frac{dI(-\tau)}{d\tau},$$

hence,

$$G(s, \tau) = -I(-\tau) + \phi(s).$$

Again, with some ϕ stationary for over constant s .

With the initial condition, $G(z, t = 0) = z(s, \tau = 0) = s$,

$$G(s, 0) = s = -I(0) + \phi(s),$$

ϕ can be fixed: $\phi(s) = 1 + s$;

$$G(s, \tau) = 1 - I(-\tau) + s.$$

Identical to the catastrophe case, the final characteristic equation is solved in the same way for the same result, (19). So finally,

$$G(z, t) = 1 - I(t) + \frac{ab(1 - e^{\beta(a-b)t}) - z(a - be^{\beta(a-b)t})}{(b - ae^{\beta(a-b)t}) - z(1 - e^{\beta(a-b)t})}.$$

And this result makes good sense. With $1 - I(t)$ being the likelihood that the process has made the “killing” transition to 0, these two terms stand to complement the probability that state 0 is reached from individual death (this component arises from the final term).

4.2 Several founders

Both solutions extend to an arbitrary initial condition by the branching property. If the cell begins with k independent founders, $X_0 = k$, then the k lineages evolve independently and the total count is their sum, so the probability generating function is the k th power of the single-founder probability generating function:

$$G_k(z, t) = (G(z, t))^k, \quad (20)$$

in either rupture convention. In the catastrophe convention this is a defective probability generating function with $G_k(1, t) = I(t)^k$; in the killing convention it remains proper. Formula (20) will be used in §9 (multiplicity of infection) and §10 (chained immediate transfer).

5 State probabilities

Using formula (16), we can obtain the individual state probabilities. The first, $p_0(t)$, differs between the catastrophe and killing cases:

$$G(0, t) = \begin{cases} \frac{ab(1 - e^{\beta(a-b)t})}{(b - ae^{\beta(a-b)t})}, & \text{(catastrophe)} \\ 1 - I(t) + \frac{ab(1 - e^{\beta(a-b)t})}{(b - ae^{\beta(a-b)t})}, & \text{(killing)}. \end{cases}$$

As stated above, the fraction term in each case represents the probability of reaching 0 by the death of every individual, and the $1 - I(t)$ pertains to reaching 0 by collapse in the killing convention.

From here, the state probabilities $p_1(t)$, $p_2(t)$, $p_3(t) \dots$ do not differ between the cases. Repeated differentiation of $G(z, t)$ with respect to z reveals the formula,

$$\frac{\partial^n G}{\partial z^n} = n! e^{\beta(a-b)t} (a-b)^2 \frac{(1 - e^{\beta(a-b)t})^{n-1}}{(b - ae^{\beta(a-b)t} - (1 - e^{\beta(a-b)t})z)^{n+1}}$$

Hence, setting $z = 0$, and dividing by $n!$, the state probabilities for $n \geq 1$ are described by,

$$p_n(t) = \left(\frac{a-b}{b - ae^{\beta(a-b)t}} \right)^2 e^{\beta(a-b)t} \left(\frac{1 - e^{\beta(a-b)t}}{b - ae^{\beta(a-b)t}} \right)^{n-1}. \quad (21)$$

5.1 Geometric at every time

There is more structure here than (21) first suggests. Writing

$$P(t) := \frac{1 - e^{\beta(a-b)t}}{b - ae^{\beta(a-b)t}}, \quad (22)$$

the entire n -dependence of $p_n(t)$ lives in the single factor $P(t)^{n-1}$:

$$p_n(t) = p_1(t) P(t)^{n-1}, \quad n \geq 1. \quad (23)$$

That is, *at every time* t , conditional on the population being non-zero, the load is geometrically distributed with ratio $P(t)$. The ratio itself evolves monotonically, with

$$P(0) = 0, \quad P(t) \xrightarrow[t \rightarrow \infty]{} \frac{1}{a}, \quad (24)$$

so the transient geometric laws slide towards a fixed limit. This limit is the heart of §6–§7.

5.2 Normalisation: the sum is $\hat{I}$

A useful consistency check is to sum (21) over the positive states. As a geometric series,

$$\sum_{n=1}^{\infty} p_n(t) = \frac{p_1(t)}{1 - P(t)}.$$

Now

$$1 - P(t) = \frac{b - ae^{\beta(a-b)t} - 1 + e^{\beta(a-b)t}}{b - ae^{\beta(a-b)t}} = \frac{-(1-b) - e^{\beta(a-b)t}(a-1)}{b - ae^{\beta(a-b)t}} = \frac{B + Ae^{\beta(a-b)t}}{ae^{\beta(a-b)t} - b},$$

and therefore

$$\sum_{n=1}^{\infty} p_n(t) = \frac{(a-b)^2 e^{\beta(a-b)t}}{(B + Ae^{\beta(a-b)t})(ae^{\beta(a-b)t} - b)} = \hat{I}(t). \quad (25)$$

The sum of the positive state probabilities is exactly the corrected non-fixation probability of §2.4 — the probability of being neither ruptured nor extinct. Equation (25) thereby confirms, by a completely independent route, the correction of Remark 2.2: the state probabilities of this chapter were already implying the right $\hat{I}(t)$.

6 Quasi-stationary distribution

[4]. Both the BDC and BDK processes are subject to eventual absorption — either at the designated rupture state, or by death, at 0. But suppose there is an example of a particular realisation which has, by chance, not yet reached fixation. It continues, increasingly against the odds, as time marches on. How has it managed this? If it flies too low, depletion to zero becomes more and more probable; but, almost as bad, if it flies high, a population booming in number, there are all the more opportunities for its downfall, a crashing and ceasing to be. Like Icarus, the process which does not fall into the sea does so by hovering at some appropriate height; its wings neither undermined by the spray of a hungry ocean, nor by the beating rays of the sun at a rarified altitude.

In such terms, we can examine this rogue process as it goes on to defy all odds. Its fate will tend to be predicted by the quasi-stationary distribution: a set of limiting conditional probabilities, each qualified by non-fixation.

6.1 Conditioning on survival

Define the probability of fixation by time t : $F(t)$, whether by death or collapse. This is settled immediately by the decomposition of §2.4: a realisation is fixed by t precisely when it is not productively infected, so

$$F(t) = 1 - \hat{I}(t) = 1 - I(t) + D(t), \quad (26)$$

which reduces to $F(t) = 1 - I(t)$ when $\mu = 0$. The conditional state probabilities decompose as

$$\Pr \{\mathbf{X}_t = i\} = \Pr \{\mathbf{X}_t = i \mid \text{fixation}\} F(t) + \Pr \{\mathbf{X}_t = i \mid \text{non-fixation}\} (1 - F(t)).$$

For $i \notin \{0, R\}$ the process has not been absorbed, so $\Pr \{\mathbf{X}_t = i \mid \text{fixation}\} = 0$, and it follows that

$$\Pr \{\mathbf{X}_t = i \mid \text{non-fixation}\} = \frac{p_i(t)}{1 - F(t)} = \frac{p_i(t)}{\hat{I}(t)}. \quad (27)$$

The *quasi-stationary distribution* (QSD) is the limiting form of these conditional probabilities:

$$q_i := \lim_{t \rightarrow \infty} \Pr \{\mathbf{X}_t = i \mid \mathbf{X}_t \notin \{0, R\}\} = \lim_{t \rightarrow \infty} \frac{p_i(t)}{\hat{I}(t)}, \quad i \geq 1. \quad (28)$$

6.2 The quasi-stationary distribution is geometric

With the state probabilities (21) in hand, the limit can be taken explicitly. By (27) and (25),

$$\Pr\{\mathbf{X}_t = n \mid \mathbf{X}_t \notin \{0, R\}\} = \frac{p_n(t)}{\sum_{m \geq 1} p_m(t)} = (1 - P(t)) P(t)^{n-1},$$

for every t . Since $P(t) \rightarrow 1/a$ by §5.1, the limit exists and is explicit.

Theorem 6.1 (Quasi-stationary distribution of the BDC). *For all $\mu \geq 0$, the quasi-stationary distribution of the BDC process is geometric on $\{1, 2, \dots\}$ with ratio $1/a$:*

$$q_n = \left(1 - \frac{1}{a}\right) \left(\frac{1}{a}\right)^{n-1} = (a-1) a^{-n}, \quad n \geq 1. \quad (29)$$

Its moments follow at once:

$$\langle X \rangle_{\text{QS}} = \frac{a}{a-1} = \frac{a}{A}, \quad \langle X^2 \rangle_{\text{QS}} = \frac{a(a+1)}{(a-1)^2}, \quad \text{Var}_{\text{QS}}(X) = \frac{a}{(a-1)^2} = \frac{a}{A^2}. \quad (30)$$

Observe that $\langle X \rangle_{\text{QS}} = \mathbb{E}(\mathcal{K} \mid \text{burst})$ from (12), and that when $\mu = 0$ both coincide with V_∞ — the coincidence noted in §2.5. For $\mu > 0$ they differ: at $(\beta, \mu, \delta) = (1, 0.2, 0.05)$, for example, $\langle X \rangle_{\text{QS}} \approx 17.23$ while $V_\infty \approx 13.99$. Section §7 will give the reason: the burst size is not merely equal in mean to the QS level — the entire burst-size distribution *is* the quasi-stationary distribution.

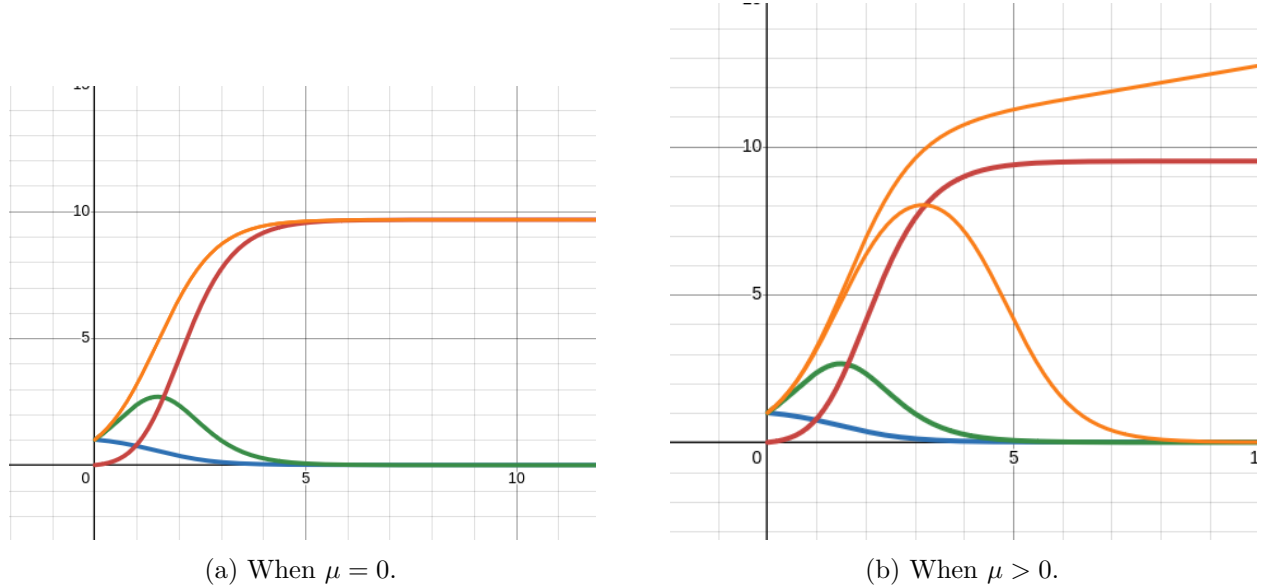


Figure 1: Conditional means from simulation. (a) When $\mu = 0$, the conditional mean $J(t)/I(t)$ tends to the same limit as $V(t)$, namely $1 + \beta/\delta$. (b) When $\mu > 0$, the curve rising at late time is the mean conditioned on non-fixation, $J(t)/\hat{I}(t)$, tending to $\langle X \rangle_{\text{QS}} = a/(a-1)$; the curve tending to 0 is the mean conditioned only on non-catastrophe, $\mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t \neq R) = J(t)/I(t)$, which vanishes because a fraction b of realisations die out internally.

Considering the no-death $\mu = 0$ case on the left above, it can be easily seen that the conditional mean tends to the same limit as $V(t)$, which is shown in red. That is

$$\lim_{t \rightarrow \infty} \frac{J(t)}{I(t)} = \lim_{t \rightarrow \infty} V(t) = 1 + \frac{\beta}{\delta}.$$

Here, when δ shrinks to 0, the QS mean and expected release count tend to infinity.

6.3 Mean productive lifetime

Before leaving the single-cell description, one more integral will be needed in §13. The expected duration of productive infection is

$$\mathbb{E}(T_{\text{prod}}) = \int_0^\infty \hat{I}(t) dt, \quad (31)$$

since $\hat{I}(t) = \mathbb{P}\{T_{\text{prod}} > t\}$.

Proposition 6.2.

$$\mathbb{E}(T_{\text{prod}}) = \frac{1}{\beta} \log\left(\frac{a}{a-1}\right) = \frac{1}{\beta} \log\langle X \rangle_{\text{QS}}. \quad (32)$$

Proof. Write $\hat{I} = (I - b) - (D - b)$. From (7),

$$I(t) - b = \frac{B(a-b)}{B + Aw}, \quad D(t) - b = -\frac{b(a-b)}{aw - b},$$

so that, with $w = e^{\theta t}$ and $dt = dw/(\theta w)$,

$$\int_0^\infty \hat{I} dt = \frac{a-b}{\theta} \int_1^\infty \left(\frac{B}{w(B + Aw)} + \frac{b}{w(aw - b)} \right) dw.$$

By partial fractions, $\frac{B}{w(B + Aw)} = \frac{1}{w} - \frac{A}{B + Aw}$ and $\frac{b}{w(aw - b)} = -\frac{1}{w} + \frac{a}{aw - b}$, hence

$$\int_1^\infty \frac{B dw}{w(B + Aw)} = \left[\log \frac{w}{B + Aw} \right]_1^\infty = \log \frac{a-b}{a-1},$$

$$\int_1^\infty \frac{b dw}{w(aw - b)} = \left[\log \frac{aw - b}{w} \right]_1^\infty = \log \frac{a}{a-b}.$$

Summing, and using $\theta = \beta(a-b)$,

$$\int_0^\infty \hat{I} dt = \frac{a-b}{\beta(a-b)} \left(\log \frac{a-b}{a-1} + \log \frac{a}{a-b} \right) = \frac{1}{\beta} \log \frac{a}{a-1}. \quad \square$$

□

When $\mu = 0$ this reduces to $\beta^{-1} \log(1 + \beta/\delta)$. The form of the answer has an interpretation worth recording: $\log\langle X \rangle_{\text{QS}}/\beta$ is the time it would take exponential growth at rate β to reach the quasi-stationary level — the productive lifetime is precisely the time needed to climb to the height at which the process can hover.

7 Burst time and burst size

The two random quantities that a bursting cell presents to the outside world are the time of its rupture and the size of the rupture. Both are now within reach of closed forms.

7.1 Burst time

The no-burst probability is $I(t) = \mathbb{P}\{\tau > t\}$, so the density of the burst time is

$$\varphi(t) := -I'(t) = \delta J(t), \quad (33)$$

using $I' = -\delta J$ from §2.5. This density is *defective*: since $I(t) \rightarrow b$,

$$\int_0^\infty \varphi(t) dt = 1 - b, \quad (34)$$

with the remaining mass b sitting at $\tau = \infty$ — the realisations that clear internally without ever bursting. Conditioned on eventual burst, the density is $\varphi(t)/(1 - b) = \delta J(t)/(1 - b)$. Where earlier work referred to “the known pdf of rupture times $\phi(t)$ ”, it is (33) that was meant, available all along from $I' = -\delta J$.

7.2 Burst size

The burst size $\mathcal{K}$ is the value of $\mathbf{X}_t$ immediately before rupture. For the burst to occur in state k , the process must be in state k and then rupture from there; rupturing from state k happens at rate δk . Hence

$$\Pr\{\mathcal{K} = k\} = \delta k \int_0^\infty p_k(t) dt. \quad (35)$$

A direct discretisation makes this plain: partitioning time into intervals of width Δt , the probability that the burst occurs during an interval in which the process is at k is

$$\Pr\{\mathbf{X}_t = k\} \Pr\{\text{rupture in } \Delta t \mid \mathbf{X}_t = k\} = \Pr\{\mathbf{X}_t = k\} \delta k \Delta t,$$

and summing over intervals and passing to the limit $\Delta t \rightarrow 0$ yields (35).

The integral is evaluable in closed form. With $P(t)$ as in (22),

$$\frac{d}{dt} \left[\frac{P(t)^k}{k\beta} \right] = \frac{1}{\beta} P(t)^{k-1} P'(t) = \left(\frac{a-b}{b - ae^{\beta(a-b)t}} \right)^2 e^{\beta(a-b)t} P(t)^{k-1} = p_k(t), \quad (36)$$

as one verifies from $P'(t) = \beta(a-b)^2 e^{\beta(a-b)t} / (b - ae^{\beta(a-b)t})^2$. Therefore, since $P(0) = 0$ and $P(\infty) = 1/a$,

$$\int_0^\infty p_k(t) dt = \frac{1}{k\beta} \left(\frac{1}{a} \right)^k, \quad (37)$$

and (35) gives the burst-size law.

Theorem 7.1 (Burst-size distribution). *Unconditionally (including the mass b of realisations that never burst),*

$$\Pr\{\mathcal{K} = k\} = \frac{\delta}{\beta} a^{-k}, \quad k \geq 1; \quad (38)$$

conditioned on the burst occurring,

$$\Pr\{\mathcal{K} = k \mid \text{burst}\} = (a-1) a^{-k}, \quad k \geq 1. \quad (39)$$

Proof. The unconditional form is the computation above. For the conditional form, divide (38) by the eventual-burst probability $1 - b$:

$$\Pr\{\mathcal{K} = k \mid \text{burst}\} = \frac{(\delta/\beta) a^{-k}}{1 - b} = \frac{AB}{B} a^{-k} = (a-1) a^{-k},$$

using $\delta/\beta = AB$ from (3). □

No simulation is required; nevertheless two consistency checks close the loop pleasantly. First, normalisation:

$$\sum_{k=1}^{\infty} \frac{\delta}{\beta} a^{-k} = \frac{\delta}{\beta} \frac{1}{a-1} = \frac{AB}{A} = B = 1 - b,$$

matching the eventual-burst probability. Second, the first moment:

$$\sum_{k=1}^{\infty} k \frac{\delta}{\beta} a^{-k} = \frac{\delta}{\beta} \frac{a}{(a-1)^2} = \frac{aB}{A} = V_{\infty},$$

recovering (12). This second check independently validates the burst-size law, the formula for V_{∞} , and the identity (3) simultaneously.

7.3 Burst size equals the quasi-stationary distribution

Compare (39) with Theorem 6.1: they are the same law.

Corollary 7.2 (Burst size = QSD). *The burst-size distribution, conditioned on bursting, is exactly the quasi-stationary distribution of the load: both are geometric on $\{1, 2, \dots\}$ with ratio $1/a$.*

This is not obvious, and it deserves a moment's pause. A burst is a *size-biased* draw from the transient load distribution, taken against the burst intensity, and then integrated over all burst times; that this should land exactly on the conditional limiting distribution is a genuine identity, not a definition. The algebra above proves it; a structural argument — presumably relating size-biasing composed with the burst intensity to the Yaglom limit [5] — would make it quotable rather than coincidental, and remains open (see §17.4).

7.4 Size-biasing and the size of a late burst

The size-biasing mechanism is worth stating on its own terms. A cell bursts at rate $\delta \mathbf{X}_t$, so given that the burst occurs at time t , its size is a size-biased draw from $p.(t)$:

$$\Pr \{ \mathcal{K} = k \mid \tau = t \} = \frac{k p_k(t)}{J(t)}, \quad \mathbb{E}(\mathcal{K} \mid \tau = t) = \frac{K(t)}{J(t)} = 1 + \frac{2\beta}{\delta} (1 - I(t)). \quad (40)$$

Taking $t \rightarrow \infty$,

$$\mathbb{E}(\mathcal{K} \mid \tau = t) \xrightarrow{t \rightarrow \infty} 1 + \frac{2\beta}{\delta} (1 - b) = \frac{\langle X^2 \rangle_{\text{QS}}}{\langle X \rangle_{\text{QS}}}, \quad (41)$$

where the second equality follows from (30). A very late burst is roughly twice the size of an average one — the signature of size-biasing a geometric law: the cells that survive long are precisely the ones that grew large, and large cells burst quickly.

8 Conditional burst means

The mean burst size and its various conditionings hide a small trap. It is worth springing it deliberately.

8.1 The overall expected burst size

Recall that $V(t) = \mathbb{E}(\mathbf{W}_t)$, where $\mathbf{W}_t$ is the number of particles released by time t . Whether the catastrophe has taken place or not can be represented by the truth of $\mathbf{W}_t > 0$, since the minimum burst size is 1; rupture has occurred with probability $1 - I(t)$. Decomposing,

$$\mathbb{E}(\mathbf{W}_t) = \mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t > 0)(1 - I(t)) + \mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t = 0)I(t),$$

and since $\mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t = 0) = 0$,

$$\mathbb{E}(\mathbf{W}_t) = \mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t > 0)(1 - I(t)),$$

that is,

$$\mathbb{E}(\mathbf{W}_t \mid \mathbf{W}_t > 0) = \frac{V(t)}{1 - I(t)}. \quad (42)$$

This can be read as “mean burst size, given that a burst has happened by t ”. The overall expected burst size — the expected size of an arbitrarily generated burst — is the late-time value of this expression:

$$\mathbb{E}(\mathcal{K}) = \lim_{t \rightarrow \infty} \frac{V(t)}{1 - I(t)} = \frac{V_\infty}{1 - b} = \frac{\beta - \mu}{\delta} + \frac{1}{1 - b}, \quad (43)$$

where the last form follows from (12), and $\mathbb{E}(\mathcal{K}) = a/(a-1)$ as already noted in (39). The distinction matters: V_∞ itself averages in the realisations that cleared internally and released nothing, so V_∞ is *not* the mean burst size unless $\mu = 0$.

8.2 The curious circularity, resolved

What is the expected burst size of a cell that has *not yet* burst? Decomposing $\mathbb{E}(\mathcal{K})$ on the event $\{\tau > t\}$,

$$\begin{aligned} \mathbb{E}(\mathcal{K}) &= \mathbb{E}(\mathcal{K} \mid \tau > t)\Pr\{\tau > t\} + \mathbb{E}(\mathcal{K} \mid \tau < t)\Pr\{\tau < t\} \\ &= \mathbb{E}(\mathcal{K} \mid \tau > t)I(t) + \frac{V(t)}{1 - I(t)}(1 - I(t)), \end{aligned} \quad (44)$$

so that

$$\mathbb{E}(\mathcal{K} \mid \tau > t) = \frac{\mathbb{E}(\mathcal{K}) - V(t)}{I(t)} = \frac{V_\infty/(1 - b) - V(t)}{I(t)}. \quad (45)$$

So far, so good. The trap is what happens when one takes $t \rightarrow \infty$ in (45). A first attempt gives

$$\lim_{t \rightarrow \infty} \mathbb{E}(\mathcal{K} \mid \tau > t) = \frac{V_\infty/(1 - b) - V_\infty}{b},$$

which, after a little algebra, appears to return $\mathbb{E}(\mathcal{K})$ itself — a curious circularity: the burst size of a cell that has waited an infinite time to burst seems to equal the burst size of a typical cell, which makes no sense, since conditioning on $\tau > t$ as $t \rightarrow \infty$ should select the realisations that never burst at all.

The resolution is that (45) mixes two conditionings. Its numerator $\mathbb{E}(\mathcal{K}) - V(t)$ is the contribution to the burst size from realisations that burst *after* t ; its denominator $I(t)$ is the probability of not having burst by t , which includes the mass b of realisations that will never burst and whose burst size is zero. Written consistently,

$$\mathbb{E}(\mathcal{K} \mathbf{1}\{\text{burst}\} \mid \tau > t) = \frac{V_\infty - V(t)}{I(t)} \xrightarrow{t \rightarrow \infty} \frac{V_\infty - V_\infty}{b} = 0, \quad (46)$$

which is correct and unmysterious: conditioning on “has not burst yet” converges to “never bursts”, on which event the burst size is zero.

The genuinely interesting limit conditions on eventual burst as well:

$$\mathbb{E}(\mathcal{K} \mid \tau > t, \text{ burst eventually}) = \frac{V_\infty - V(t)}{I(t) - b}. \quad (47)$$

This is of the form $0/0$ as $t \rightarrow \infty$; by l'Hôpital's rule, using $V' = \delta K$ and $I' = -\delta J$,

$$\lim_{t \rightarrow \infty} \mathbb{E}(\mathcal{K} \mid \tau > t, \text{ burst eventually}) = \lim_{t \rightarrow \infty} \frac{K(t)}{J(t)} = \frac{\langle X^2 \rangle_{\text{QS}}}{\langle X \rangle_{\text{QS}}} = 1 + \frac{2\beta}{\delta}(1 - b), \quad (48)$$

in agreement with (40): the eventual burst size of a very patient cell is the size-biased quasi-stationary mean. The circularity was never in the mathematics; it was in the conditioning.

9 Multiplicity of infection

So far the cell has been founded by a single particle. In applications this is not always the right picture: a macrophage may be invaded by several bacteria at once, and in the laboratory the multiplicity of infection is a control parameter. Nothing essential changes — the process is still linear — but the bookkeeping does, and one plausible guess turns out to be wrong. Throughout this section, k founders share a single cell and hence a single catastrophe clock: rupture occurs at rate δ times the *total* load, and dumps the whole cell.

9.1 Fixation functions for k founders

By the branching property, the k founder lineages evolve independently until the shared clock rings. No burst occurs by t precisely when no lineage has burst; internal extinction occurs precisely when every lineage has cleared internally. Hence

$$I_k(t) = I(t)^k, \quad D_k(t) = D(t)^k, \quad \hat{I}_k(t) = I(t)^k - D(t)^k. \quad (49)$$

Remark 9.1 (A guess that fails). The tempting guess is $\hat{I}_k = \hat{I}^k$: each lineage stays productive with probability $\hat{I}$, independently. But “productive” for the whole cell means *not fixed*, and the intersection of the independent events “lineage i avoids bursting and does not go extinct” is not the event “the cell is not fixed”: the cell also stops being productive when every lineage dies out. Correctly, $\hat{I}_k = I^k - D^k$, whereas $\hat{I}^k = (I - D)^k$; already for $k = 2$ the two differ by $I^2 - D^2 - (I - D)^2 = 2D(I - D) \neq 0$ whenever $\mu > 0$. Numerically, at $(\beta, \mu, \delta) = (1, 0.2, 0.05)$ and $t = 1$: the correct $\hat{I}_2 = I^2 - D^2 \approx 0.8458$, while $\hat{I}^2 \approx 0.6542$. When $\mu = 0$, $D \equiv 0$ and the two coincide — again the invisible case.

9.2 Moments and the release kernel

The moments fall out of the k -founder probability generating function of §4.2, $G_k = G^k$, where G is the single-founder (defective) probability generating function. Differentiating at $z = 1$, with $G(1, t) = I(t)$, $\partial_z G|_{z=1} = J(t)$ and $\partial_z^2 G|_{z=1} = K(t) - J(t)$,

$$J_k(t) = k I(t)^{k-1} J(t), \quad (50)$$

and, from the second factorial moment,

$$\begin{aligned}
K_k(t) &= \partial_z^2 G_k|_{z=1} + \partial_z G_k|_{z=1} \\
&= k(k-1)I(t)^{k-2}J(t)^2 + kI(t)^{k-1}(K(t) - J(t)) + kI(t)^{k-1}J(t) \\
&= kK(t)I(t)^{k-1} + k(k-1)J(t)^2I(t)^{k-2}.
\end{aligned} \tag{51}$$

The expected release flux — the population quantity that will build the renewal model of §12 — is therefore

$$g_k(t) = \delta K_k(t) = \delta \left(kK(t)I(t)^{k-1} + k(k-1)J(t)^2I(t)^{k-2} \right). \tag{52}$$

Remark 9.2 (Correction of the free-sum form). An earlier note gave $g_k = \delta(kK + k(k-1)J^2)$, the second moment of a free sum of k independent copies. That omits the powers of I : the second-moment contribution of a configuration is counted only while the shared clock has *not* rung. At $(\beta, \mu, \delta) = (1, 0.2, 0.05)$, $t = 1$, $k = 2$: the free-sum form gives $K_2 \approx 23.39$; the corrected (51) gives $K_2 \approx 22.264$, which agrees with a direct master-equation computation. The difference is the superlinearity of g_k in k wearing a disguise: even the free-sum form is superlinear, but the true kernel is superlinear *and* tempered by survival. That a cell's expected release flux depends superlinearly on its founding multiplicity is a genuine prediction of the model — one that classical constant- p models cannot express, and that low-multiplicity single-cell assays can test.

9.3 Lifetime yield from k founders

The lifetime mean release from k founders is $V_\infty^{(k)} = \int_0^\infty g_k(t) dt$. Carrying out the integration (deferred to Appendix C),

$$V_\infty^{(1)} = V_\infty = \frac{a(1-b)}{a-1}, \tag{53}$$

and, for $k \geq 2$,

$$V_\infty^{(k)} = (1-b^k) + \frac{2\beta}{\delta} \left[\frac{1}{k+1} - b^k + \frac{k b^{k+1}}{k+1} \right] + k(k-1) \frac{\beta}{\delta} \int_1^b (x-a)(x-b) x^{k-2} dx. \tag{54}$$

In the no-death case the answer simplifies dramatically:

$$V_\infty^{(k)} = k + \frac{\beta}{\delta} \quad (\mu = 0). \tag{55}$$

Indeed, when $\mu = 0$ the event type is independent of the load (§10), so the number of births before the single catastrophe is $\text{Geom}_0(s)$ with mean $\rho/s = \beta/\delta$ whatever the founding load, and the total release is simply k founders plus that many births. For example, with $\beta = 1$, $\delta = 0.1$: $V_\infty^{(k)} = 11, 12, 13, 14, 15, \dots$

Two biological points are worth pausing on. First, the yield is *not* kV_∞ : the first catastrophe dumps the entire cell once, so additional founders add their own expected descendants but not another cell's worth of release. At $(\beta, \mu, \delta) = (1, 0.2, 0.05)$: $V_\infty \approx 13.99$, so a naive scaling would predict $2V_\infty \approx 27.97$ for two founders; the true value is $V_\infty^{(2)} \approx 17.43$. Second, while the unconditional yield grows sub-extensively in k , the *conditional* mean burst size given eventual burst, $V_\infty^{(k)}/(1-b^k)$, *increases* with k (17.23, 18.07, 19.02, 21.00 for $k = 1, 2, 3, 5$ at the same parameters): multiply infected cells are more likely to burst at all, and when they do, they burst bigger.

10 Chained immediate transfer

10.1 Setup

We now couple several cells together in the most austere way possible. Take M host cells, all empty save one, which holds a single particle. Each cell runs the $\mu = 0$ birth–catastrophe process of the previous sections (birth βn , catastrophe δn); on rupture, however, the released load does not enter an extracellular pool. It is transferred immediately and wholesale into a fresh cell, where a new BDC process begins; and so on, until every cell has ruptured. There is no extracellular phase, no clearance, and no re-infection — only a chain of bursts.

The motivation is twofold. First, founding censuses can be remarkably small: in plague, say, colony-forming units are counted in the hundreds against a vast population of macrophages, so a serial, one-cell-at-a-time picture is not absurd as a caricature of early pathogenesis. Second, the model is a tractable laboratory for multi-cell burst statistics — the sizes and times of successive ruptures — which interpolate between the single-cell theory above and the population models of Part II. Since $\mu = 0$, internal extinction is impossible and all M ruptures occur almost surely.

Let r_i be the load at the i th rupture (the i th *rupture size*), t_i its time, and $T_i = t_i - t_{i-1}$ the i th inter-rupture interval (with $t_0 = 0$). Write

$$s = \frac{\delta}{\beta + \delta}, \quad \rho = \frac{\beta}{\beta + \delta} = 1 - s. \quad (56)$$

10.2 The key observation: event types ignore the load

When the load is n , the next event is a birth with probability $\beta n / (\beta n + \delta n) = \rho$ and a catastrophe with probability s — independent of n . Only the holding time depends on the load: it is $\text{Exp}((\beta + \delta)n)$. Consequently the sequence of event *types* is a sequence of independent coin flips, and the number of births G that occur before the catastrophe in a cell founded with any load is

$$G \sim \text{Geom}_0(s), \quad \Pr\{G = g\} = s \rho^g, \quad g = 0, 1, 2, \dots \quad (57)$$

(the number of failures before the first success). Each cell in the chain contributes an *independent* copy G_i of this variable — the Markov process restarts afresh at every transfer — and the rupture size of cell i is its founding load plus its own births. Since the first cell is founded by a single particle,

$$r_k = 1 + G_1 + \dots + G_k, \quad (58)$$

with $G_1, \dots, G_k$ i.i.d. $\text{Geom}_0(s)$. Everything about the rupture sizes follows from this decomposition.

10.3 Rupture-size laws

The sum of k i.i.d. geometric variables is negative binomial, so (58) gives immediately:

$$\Pr\{r_k = n\} = \mathcal{A}_k^{(n)} s^k \rho^{n-1}, \quad \mathcal{A}_k^{(n)} = \binom{n+k-2}{k-1}, \quad n \geq 1. \quad (59)$$

The first few cases display the pattern:

$$\Pr\{r_1 = n\} = s \rho^{n-1} \text{ (geometric — matches one-cell BDC)}, \quad (60)$$

$$\Pr\{r_2 = n\} = n s^2 \rho^{n-1}, \quad \Pr\{r_3 = n\} = \frac{n(n+1)}{2} s^3 \rho^{n-1}, \quad \Pr\{r_4 = n\} = \frac{n(n+1)(n+2)}{6} s^4 \rho^{n-1}. \quad (61)$$

Equivalently, the coefficients satisfy the recurrence

$$\mathcal{A}_k^{(n)} = \sum_{i=1}^n \mathcal{A}_{k-1}^{(i)}, \quad \mathcal{A}_1^{(i)} = 1, \quad (62)$$

which is just (59) conditioned on the value of r_{k-1} : $\Pr\{r_k = n\} = \sum_i \Pr\{r_{k-1} = i\} \Pr\{G_k = n - i\}$. The closed form $\mathcal{A}_k^{(n)} = \binom{n+k-2}{k-1}$ then follows by induction from the hockey-stick identity — supplying the proof that the original investigation had conjectured but not completed.

Remark 10.1 (Correction of the draft's closed form). The draft recorded $\mathcal{A}_k^{(n)} = \frac{1}{k!} \frac{(n+k-1)!}{(n-1)!} = \binom{n+k-1}{k}$, a formula off by one shift: it contradicts the draft's own r_2 and r_3 cases, which (59) reproduces exactly. The correct coefficient is $\binom{n+k-2}{k-1}$.

Further consequences, all verified numerically (see §10.7):

$$\mathbb{E}(r_k) = 1 + k \frac{\rho}{s} = 1 + k \frac{\beta}{\delta}, \quad \text{Var}(r_k) = k \frac{\rho}{s^2}, \quad (63)$$

$$\mathbb{E}(z^{r_k}) = z \left(\frac{s}{1 - \rho z} \right)^k, \quad \sum_{n \geq 1} \Pr\{r_k = n\} = 1 \text{ (burst is certain for } \mu = 0\text{)}. \quad (64)$$

The mean rupture size grows linearly along the chain — each cell adds one expected burst's worth of births, β/δ , to whatever it inherited.

10.4 Rupture times

The first rupture time inherits the single-cell flux. With m founders, the no-rupture probability is $I(t)^m$ by (49) (with $\mu = 0$, $D \equiv 0$), so

$$\Pr\{t_1 > t\} = I(t)^m, \quad f_{t_1}^{(m)}(t) = -m I(t)^{m-1} I'(t) = m I(t)^{m-1} \delta J(t). \quad (65)$$

For $m = 1$ this is the burst-time density $\varphi = \delta J$ of §7.1 (which is proper here, since $b = 0$). When $\mu > 0$, the same reasoning distinguishes the two fixation routes, as in the original investigation: the *burst-only* time has CDF $(1 - I(t))/(1 - b)$ and density $\delta J(t)/(1 - b)$, while the *general-fixation* time — rupture or internal extinction — has density

$$\delta J(t) + \mu p_1(t), \quad (66)$$

the extra term being the flux through $1 \rightarrow 0$: only a population of size exactly one can die out in a single death event.

10.5 Inter-rupture intervals

Given k founders, the interval $T(k)$ until rupture decomposes over the holding times: if $G = g$ births precede the catastrophe, the load runs through $k, k+1, \dots, k+g$, so

$$T(k) = \sum_{j=k}^{k+G} \xi_j, \quad \xi_j \sim \text{Exp}((\beta + \delta)j) \text{ independent}. \quad (67)$$

Averaging over $G \sim \text{Geom}_0(s)$, the Laplace transform is

$$\tilde{T}(k, u) = \mathbb{E}(e^{-uT(k)}) = \sum_{g \geq 0} s \rho^g \prod_{j=k}^{k+g} \frac{(\beta + \delta)j}{(\beta + \delta)j + u}. \quad (68)$$

Writing $u' = u/(\beta + \delta)$ and unrolling the product with Gamma functions, the series sums to a hypergeometric form, continuing the theme of §13:

$$\tilde{T}(k, u) = \frac{s k}{k + u'} {}_2F_1(k + 1, 1; k + 1 + u'; \rho). \quad (69)$$

Differentiating (68) at $u = 0$ gives the exact mean

$$\mathbb{E}(T(k)) = \sum_{m \geq 0} \frac{\rho^m}{(\beta + \delta)(k + m)}, \quad (70)$$

which is strictly decreasing in k : the more a cell inherits, the faster it bursts. This quantifies the original investigation's qualitative observation that the chain is “partitioned by reducing intervals of time”. For $(\beta, \delta) = (1, 1)$: $\mathbb{E}(T(1..4)) \approx 0.693, 0.386, 0.273, 0.212$ for fixed founding loads; the chain intervals T_k , whose founding loads are random, decrease faster still (0.691, 0.500, 0.377, 0.292 in simulation).

10.6 The second rupture time

The chain's dependence structure is now visible. The second rupture time is $t_2 = t_1 + T_2$, but t_1 and T_2 are *not* independent: both are functions of G_1 , the number of births in the first cell. Conditional on $G_1 = g$, however, the two pieces involve disjoint sets of exponential holding times, so their Laplace transforms factor:

$$\mathbb{E}(e^{-ut_2} \mid G_1 = g) = P_1(1 + g, u) \tilde{T}(1 + g, u), \quad P_1(m, u) = \prod_{j=1}^m \frac{(\beta + \delta)j}{(\beta + \delta)j + u}. \quad (71)$$

Averaging over $G_1 \sim \text{Geom}_0(s)$:

$$\tilde{t}_2(u) = \sum_{g \geq 0} s \rho^g P_1(1 + g, u) \tilde{T}(1 + g, u). \quad (72)$$

This resolves the question with which the original investigation closed. The convolution guessed there, $\rho(t_2) = \int f(\tau) \sum_k \Pr\{r_1 = k \mid \tau\} \phi_k(t_2 - \tau) d\tau$, is the same statement in different clothes — and, as suspected, it is not the tractable one. The Laplace form (72) is; so is the density itself, computable exactly as a mixture of hypoexponential densities ($t_1 \mid G_1$ and $T_2 \mid (G_1, G_2)$ are sums of independent exponentials) and verified against simulation in §10.7. Moments assemble from the pieces, e.g. $\mathbb{E}(t_2) = \mathbb{E}(T(1)) + \sum_{n \geq 1} \Pr\{r_1 = n\} \mathbb{E}(T(n))$.

10.7 Verification record and scope

Every quantitative claim of this section was checked by exact simulation of the chained process (the event-type decomposition makes sampling direct): the suite `verify_chained_transfer.py` comprises 28 checks across three parameter sets — the coefficient identity (exact integer arithmetic); the rupture-size laws for $k = 1..4$ against 10^6 chains; moments (63); the probability generating

function (64); the t_1 density (65) vs δJ ; the interval transform (68) for fixed founders $k \in \{1, 2, 3, 5\}$; the random-founder mixture consistency of the chain intervals; the t_2 transform (72); the exact hypoexponential-mixture density of t_2 (relative $L^2 \approx 3\%$ against histograms); and the monotone interval means. **Final run 2026-08-07: 28/28 PASS.**

What remains open, and is stated plainly: the joint laws of (t_k, r_k) for general $\mu > 0$ (the Θ/Ω coefficient machinery of the draft, with its ${}_2F_1$ antiderivatives, is the route; see Appendix C), and the joint law of (t_M, r_M) for the full M -cell chain. For $\mu = 0$, the decomposition (58) extends the size laws to all M at once, since nothing about it depends on where a cell sits in the chain.

11 From one cell to a population: why a constant release rate fails

The single-cell theory is complete. We now ask what it implies at the level of a population of infected cells — and begin with the standard object it is meant to improve upon.

11.1 The classical BMVR as comparator

In the simplified (target-cell-constant) Basic Model of Viral Replication [6, 7, 8], infected cells $\mathcal{I}$ and free particles $\mathcal{V}$ obey

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad (73)$$

with target cells T held fixed, infection rate γ , virion clearance c , and where the two constants that concern us are p , the per-cell release rate, and $d_{\mathcal{I}}$, the infected-cell removal rate. Both are phenomenological: they are parameters to be fitted, not quantities derived from anything happening inside the cell. The basic reproduction number of (73) is

$$R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}. \quad (74)$$

Every cell, regardless of its age or its contents, releases at the same rate p and dies at the same rate $d_{\mathcal{I}}$. For a budding virus this is at least a plausible picture. For a lytic pathogen it is plainly false: nothing is released until the cell ruptures, and then everything is released at once. The question of this part of the chapter is what (73) should be replaced with.

11.2 The synchronous cohort

Consider first a clean special case: N_0 cells, all infected at the same instant, each running an independent copy of the single-cell process, with no reabsorption or extracellular replication. The number still productively infected at time t is $N_0 \hat{I}(t)$ (exactly in expectation, and with vanishing relative error as N_0 grows), and the expected cumulative extracellular count is simply

$$\mathcal{V}(t) = N_0 V(t), \quad (75)$$

since $V(t)$ is the expected release from one cell. In this synchronous picture, the release *rate* is $N_0 V'(t) = N_0 \delta K(t)$: it depends on the age of the cohort, and it is not constant. The classical model's $p\mathcal{I}$ can mimic such a cohort only by accident, and only briefly.

11.3 The proposal that cannot work

The natural thought — and the one originally pursued — is to replace the constant p in (73) by a constant built from the intracellular theory: the quasi-stationary mean,

$$p \stackrel{?}{=} \langle X \rangle_{\text{QS}} = \frac{a}{a-1}. \quad (76)$$

The intuition is that a long-lived infected cell hovers at the QS level, so its bursting should look like release at rate $\langle X \rangle_{\text{QS}}$. The obstruction is dimensional: p has units of particles per unit time, while $\langle X \rangle_{\text{QS}}$ is a pure count. Substituting (76) introduces a spurious factor of time into the model. Worse, when $\mu = 0$ one has $\langle X \rangle_{\text{QS}} = V_\infty$ exactly, so the proposal substitutes the *total lifetime output* of a cell in place of its *output rate*.

Two repairs suggest themselves, and both fail for related reasons. First, average the burst size over the burst-time density $\varphi = \delta J$ of §7.1:

$$\int_0^\infty \varphi(t) \mathbb{E}(K \mid \tau = t) dt = \delta \int_0^\infty K(t) dt = V_\infty, \quad (77)$$

using (40) and $V' = \delta K$ — again a count, not a rate. Second, weight the load-dependent flux by the burst-time density, $\delta \int \varphi(t) \mathbb{E}(\mathbf{X}_t^2 \mid \cdot) dt$: this has the right dimensions, but it averages over the wrong measure. The burst-time density describes *when a single cell bursts*; what a population release rate needs is an average over the *ages of the cells currently infected*. In an asynchronous infection those two distributions are not the same, and no reweighting of a single-cell count by a single-cell density can fix the substitution, because the difficulty is not in the single-cell theory at all. It is in the population bookkeeping.

11.4 What the attempt got right

Strip the failed proposal back and something correct remains. Suppose a single infected macrophage founds a chain of infections: when it ruptures, each released particle goes on to found a new intracellular process in a new cell; these in turn end at different times, and the cycles continue, beginning and ending, beginning and ending. After a sufficient time there emerges a kind of stationarity: any given release event exhibits a burst time and a burst size distributed according to the laws of §7, and short-duration processes occur more frequently than long ones simply because they turn over faster. What we want is the overall rate of release events and their sizes — the quantity that should replace the $p\mathcal{I}$ term of (73).

Two observations from §11.2–§11.3 organise what follows. First, the expected release flux of a cell of age a is not a constant but $\delta K(a)$ — the burst rate $\delta \mathbf{X}_t$ times the size $\mathbf{X}_t$, averaged. Second, at any host time the infected population is a mixture of ages, one cohort for each past moment of infection. The correct population model must therefore keep track of *cell age*: how many cells were infected when, how many of them are still productive, and how much each age class is releasing. That is precisely an age-structured, or renewal, formulation, and because both of its ingredients already exist in closed form, it is exactly solvable. We take it up next.

12 Burst-aware renewal viral dynamics

12.1 The two kernels

Age-structured population dynamics needs exactly two single-cell ingredients: how long a cell stays productive, and how much it releases at each age. Both are already in closed form.

The *survival kernel* is the probability that a cell infected a time units ago is still productively infected:

$$S(a) := \hat{I}(a) = \frac{(a-b)^2 e^{\theta a}}{(B + Ae^{\theta a})(ae^{\theta a} - b)}. \quad (78)$$

The *release kernel* is the expected release flux at cell-age a :

$$g(a) := \delta K(a) = V'(a). \quad (79)$$

The reasoning behind (79) is the one that the original attempt stated and then failed to use (§11.3): a cell bursts at rate $\delta \mathbf{X}_t$ and, when it does, releases $\mathbf{X}_t$ particles, so the expected number of particles released in $[a, a + da]$ is $\delta \mathbb{E}(\mathbf{X}_t^2) da = \delta K(a) da$. With K from (9), both kernels are explicit functions of age — this is what it means for the population model to be *forced by* the intracellular process (Figure 2).

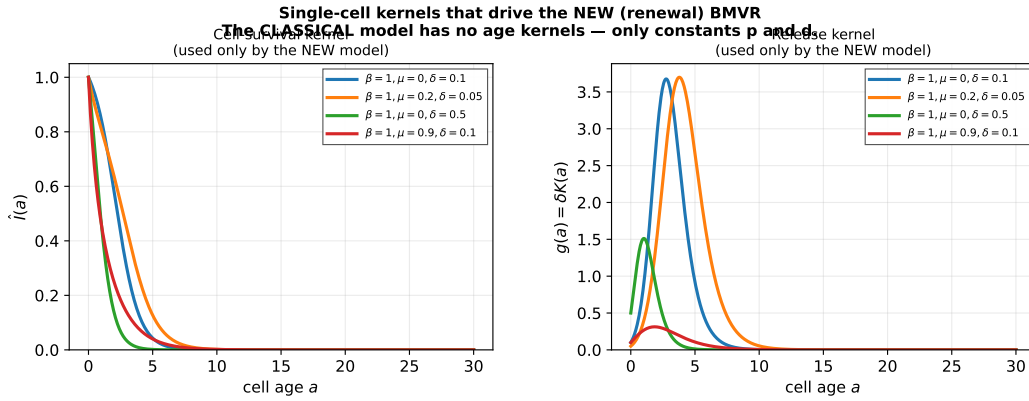


Figure 2: The cell-survival kernel $\hat{I}(a)$ and the release kernel $g(a) = \delta K(a)$ for four intracellular parameter sets. These age-dependent kernels drive the renewal model of this section; the classical model (73) has no kernels, only the constants p and d_I .

12.2 The renewal system

Hold target cells T fixed, let infection occur at rate γ per virion, clear virions at rate c , and define the incidence

$$i(t) = \gamma T \mathcal{V}(t). \quad (80)$$

Each infection event at time $t - \alpha$ starts a new cell of age α ; such a cell is still productive with probability $\hat{I}(\alpha)$ and releases at expected rate $g(\alpha)$. Summing over all infection times, and adding the contribution $\mathcal{I}_0$ of cells already infected at $t = 0$,

Theorem 12.1 (New viral dynamics — burst-aware BMVR).

$$\mathcal{I}(t) = \mathcal{I}_0 \hat{I}(t) + \int_0^t i(t - \alpha) \hat{I}(\alpha) d\alpha = \mathcal{I}_0 \hat{I} + (i * \hat{I})(t), \quad (81)$$

$$\frac{d\mathcal{V}}{dt}(t) = \mathcal{I}_0 g(t) + \int_0^t i(t - \alpha) g(\alpha) d\alpha - c \mathcal{V}(t) = \mathcal{I}_0 g + (i * g)(t) - c \mathcal{V}(t). \quad (82)$$

The correspondence with the classical model is:

Classical BMVR (73)	New system (81)–(82)
$\mathcal{I}' = i - d_{\mathcal{I}}\mathcal{I}$ (constant removal rate $d_{\mathcal{I}}$)	$\mathcal{I} = \mathcal{I}_0\hat{I} + i * \hat{I}$ (age-structured survival via $\hat{I}$)
$\mathcal{V}' = p\mathcal{I} - c\mathcal{V}$ (constant release rate p)	$\mathcal{V}' = \mathcal{I}_0g + i * g - c\mathcal{V}$ (age-structured release via $g = \delta K$)

Both constant terms are replaced, not just the release: the memoryless removal $d_{\mathcal{I}}\mathcal{I}$ becomes the age structure implicit in (81), and the constant release $p\mathcal{I}$ becomes the convolution with g . The original attempt of Chapter 4 only ever proposed to change the second.

Differentiating (81) makes the structure visible:

$$\frac{d\mathcal{I}}{dt} = i(t) + \mathcal{I}_0\hat{I}'(t) + \int_0^t i(t-\alpha)\hat{I}'(\alpha)d\alpha, \quad (83)$$

in which the incidence appears as a source exactly as in the classical model, while the remaining terms encode age-dependent loss of productivity. There is, in general, no constant $d_{\mathcal{I}}$ for which this reduces to $\mathcal{I}' = i - d_{\mathcal{I}}\mathcal{I}$ for every history of i : such a reduction holds only when everything grows exponentially, and that exception is exactly what §13 exploits.

Remark 12.2 (What is and is not new). The renewal, or age-structured, framework itself is standard: it is the McKendrick–von Foerster machinery of mathematical demography [9, 10], and age-structured variants of within-host models are well studied. The novelty here is not the framework. It is that the kernels $\hat{I}$ and g are *derived in closed form from a mechanistic intracellular stochastic process*, rather than posited phenomenologically — together with the consequences drawn in the next three sections. Any honest write-up must say so plainly.

12.3 Numerical verification

Result 12.1 was verified end-to-end by a dedicated numerical suite (`verify_result_20_1.py`); the full catalogue is Appendix B. The tests cover four layers: kernel integrity (the inputs obey the single-cell identities, five parameter sets); the $\gamma T = 0$ reduction (with no reinfection, (81)–(82) collapse to $\mathcal{I} = \mathcal{I}_0\hat{I}$ and the forced linear equation $\mathcal{V}' = \mathcal{I}_0g - c\mathcal{V}$, matching the single-cell convolution to machine precision, and with $c = 0$ giving $\mathcal{V} = \mathcal{I}_0V(t)$); structural limits (exponential kernels reproduce classical BMVR to relative error $< 5 \times 10^{-5}$, Figure 3; the late exponential growth rate equals the characteristic root r_* of §13 to $\sim 10^{-7}$, Figure 11a; the R_0 threshold is sharp, Figure 11b); and stochastic ground truth — an independent Gillespie ensemble of BDC cells with virion clearance reproduces the renewal means for $\mathcal{I}$ and $\mathcal{V}$ within sampling error ($\gamma T = 0$), Figure 4. **Headline: all 54 checks pass** (the full catalogue, with tallies and reproduction instructions, is Appendix B).

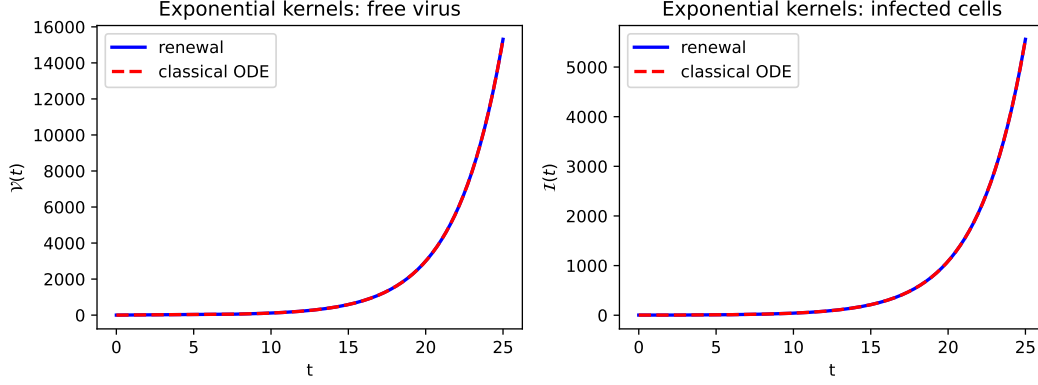


Figure 3: Structural correctness: when the kernels are pure exponentials, $S = e^{-da}$, $g = pe^{-da}$, the renewal system (81)–(82) is classical BMVR with those (p, d) . Relative discrepancy $< 5 \times 10^{-5}$ for three matched parameter pairs.

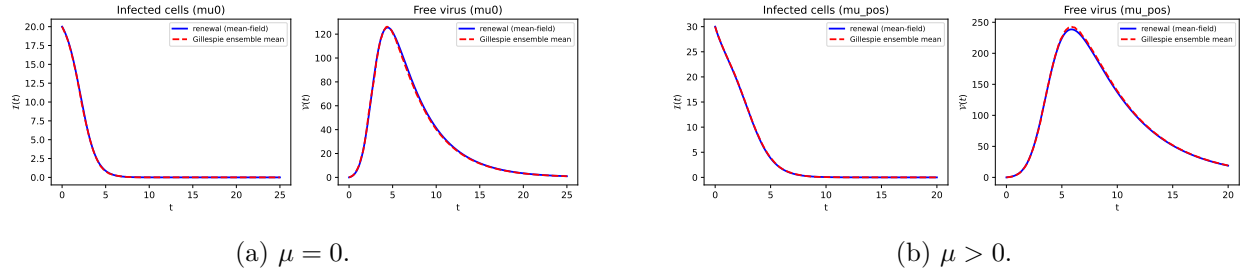


Figure 4: Stochastic ground truth: ensemble means of an independent Gillespie simulation of BDC cells (impulse releases at burst times, exponential clearance, no reinfection) against the renewal solution, relative $L^2 \sim 1\text{--}2\%$.

12.4 Overlays: classical vs renewal

To see what age structure does to an epidemic, integrate both systems side by side. In Figures 5–9, solid navy is the renewal model (81)–(82); dashed orange with circles is classical BMVR with the fairest constant-parameter match at the $r = 0$ end of the effective-rate map of §13,

$$p = p_{\text{eff}}(0) = \frac{V_{\infty}}{\mathbb{E}(T_{\text{prod}})} = \frac{\beta V_{\infty}}{\log(a/(a-1))}, \quad d_I = d_{I,\text{eff}}(0) = \frac{\beta}{\log(a/(a-1))}, \quad (84)$$

so that total lifetime output and mean productive lifetime agree.

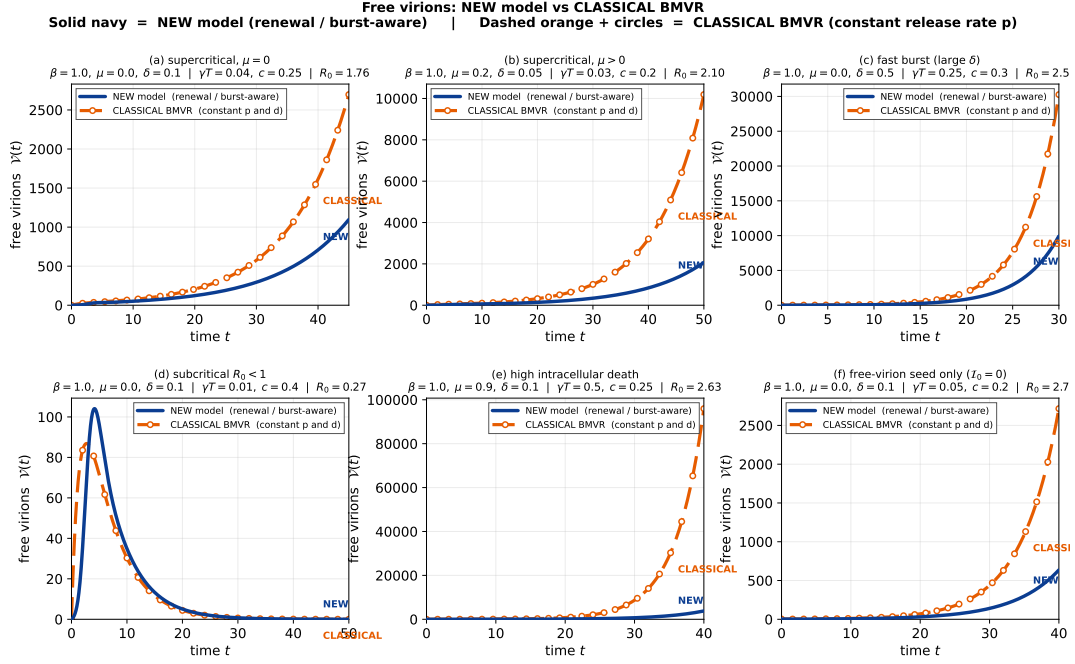


Figure 5: Free-particle trajectories $\mathcal{V}(t)$. Solid navy: renewal; dashed orange with circles: classical BMVR matched at $r = 0$. Six scenarios: supercritical growth, subcritical decay, fast burst, high intracellular death, and free-particle-only seeding.

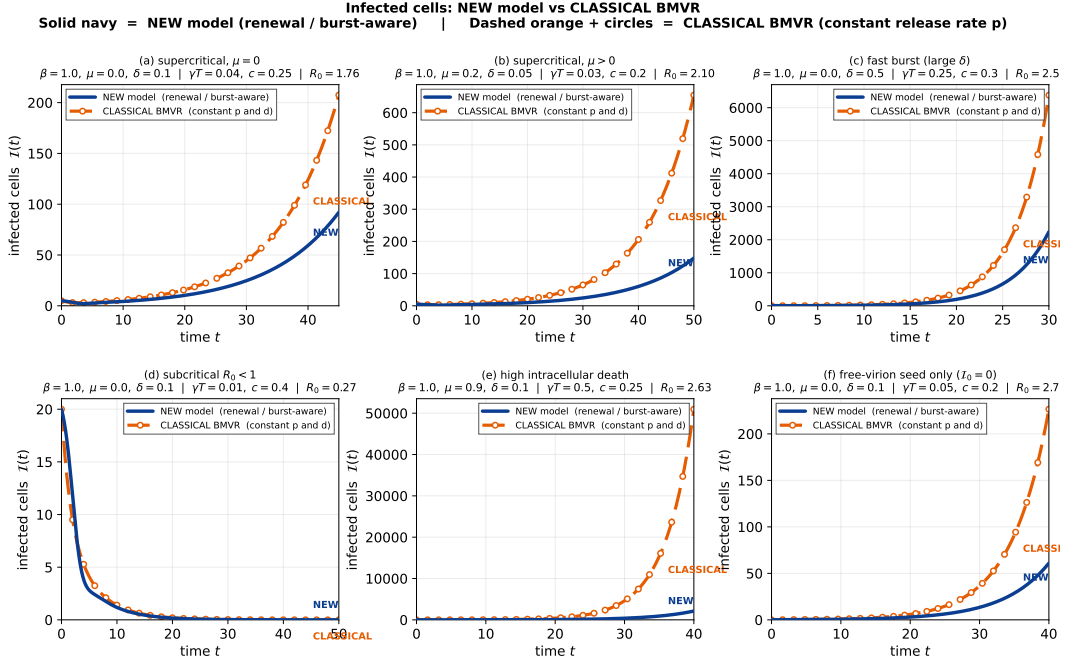


Figure 6: Infected-cell trajectories $\mathcal{I}(t)$ for the same six scenarios; same line code as Figure 5.

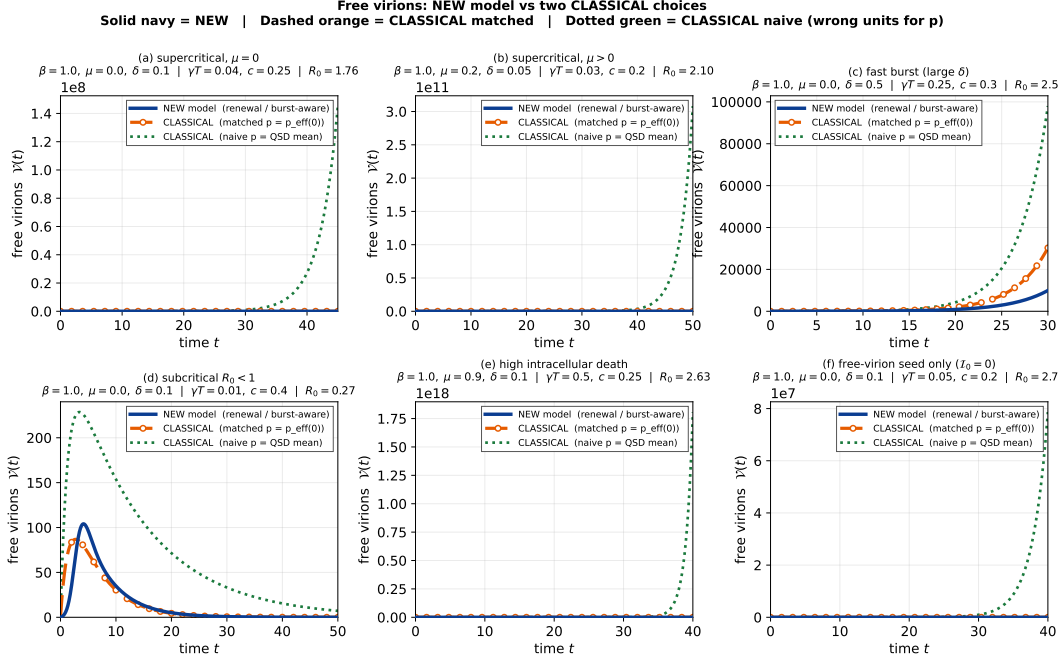


Figure 7: The naive proposal (76) as a third curve (dotted green): classical BMVR with $p = \langle X \rangle_{QS}$ — a count used as a rate — is systematically wrong in scale, not merely in shape.

The reading is consistent across panels. The matched classical model often gets the order of magnitude but not the shape: it misses the early lag (young cells dominate, and release is closer to the young-cell limit $p \rightarrow \delta$ than to $p_{\text{eff}}(0)$), the timing of the peak, and the late growth rate (as the epidemic grows, generation-time structure moves the operative release rate away from its $r = 0$ value). The relative free-particle discrepancy,

$$\frac{|\mathcal{V}^{\text{new}}(t) - \mathcal{V}^{\text{classical}}(t)|}{\max_s \mathcal{V}^{\text{new}}(s)}, \quad (85)$$

is often order one or larger by the end of the plotted window in supercritical regimes, and is worst when intracellular death is high (Figure 8): there, V_∞ is small while the age-structure of release remains strongly non-exponential. Discrepancies grow with R_0 and with the strength of age-structure in the kernels (slow bursting, non-zero μ).

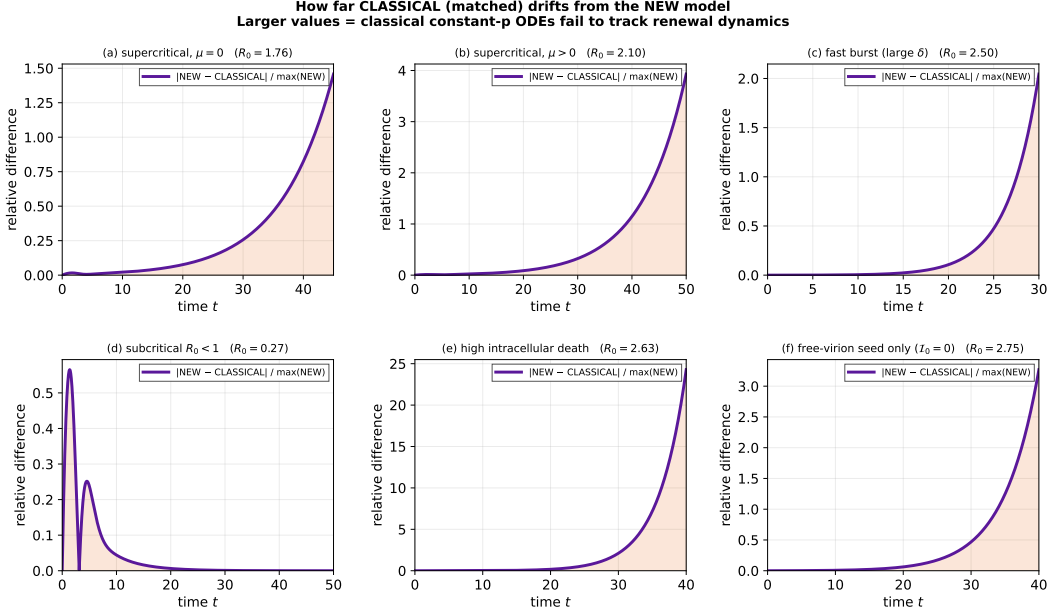


Figure 8: Relative free-particle discrepancy between the renewal model and matched classical BMVR across the six scenarios. Large values mean the constant- p ODEs fail to track the burst-aware model.

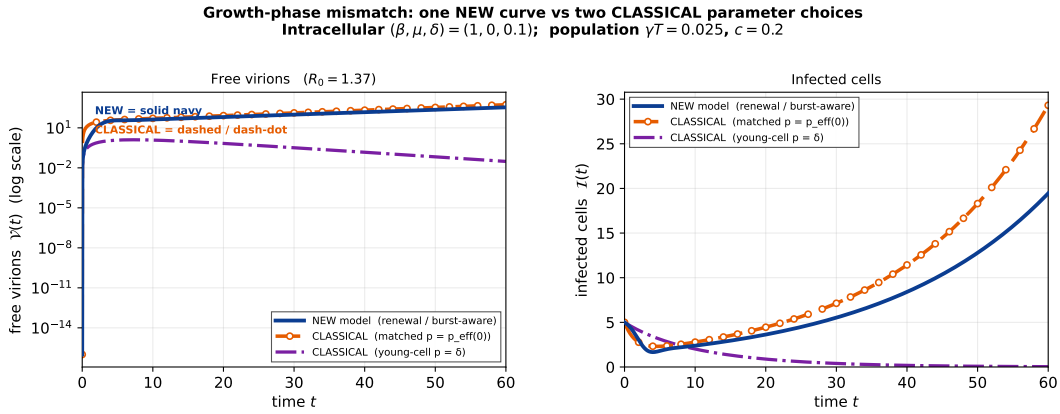


Figure 9: Growth-phase mismatch for one supercritical set $(\beta = 1, \mu = 0, \delta = 0.1)$. Solid navy: renewal; dashed orange: classical matched at $r = 0$; dash-dot purple: classical with the young-cell parameters $p = \delta$, $d_I = \mu + \delta$. Neither classical choice tracks the renewal curve across the whole trajectory — the $r = 0$ match fails early, the young-cell match fails late. Left panel logarithmic in $\mathcal{V}$.

These numerical facts are exactly what the analytic reduction of the next section predicts: a constant pair (p, d_I) can reproduce one exponential growth rate exactly, but not a transient that samples a continuum of cell ages.

13 Effective parameters, R_0 , and identifiability

The renewal model (81)–(82) does not merely disagree with classical BMVR in transients; it contains classical BMVR as an exact special case, and the relationship between the two is precise enough to be turned into predictions.

13.1 Exact reduction in the exponential phase

Suppose the incidence grows exponentially, $i(t) = i_0 e^{rt}$, and look at times late enough that the lower limits of the convolutions are irrelevant. Then

$$\mathcal{I}(t) = i_0 e^{rt} \widetilde{I}(r), \quad \mathcal{V}'(t) = i_0 e^{rt} \delta \widetilde{K}(r) - c \mathcal{V}(t), \quad (86)$$

where $\widetilde{f}(r) = \int_0^\infty e^{-r\alpha} f(\alpha) d\alpha$. A classical model $\mathcal{I}' = i - d_{\mathcal{I}} \mathcal{I}$, $\mathcal{V}' = p \mathcal{I} - c \mathcal{V}$ driven by the same incidence has $\mathcal{I}(t) = i_0 e^{rt} / (r + d_{\mathcal{I}})$ and the eigenvalue relation $(r + d_{\mathcal{I}})(r + c) = \gamma T p$. Matching the two exponential solutions fixes

$$p_{\text{eff}}(r) = \frac{\delta \widetilde{K}(r)}{\widetilde{I}(r)}, \quad d_{\mathcal{I}, \text{eff}}(r) = \frac{1}{\widetilde{I}(r)} - r. \quad (87)$$

This reduction is *exact*, not approximate: substituting (87) into $(r + d_{\mathcal{I}})(r + c) = \gamma T p$ returns

$$r + c = \gamma T \delta \widetilde{K}(r), \quad (88)$$

which is precisely the characteristic equation of the renewal system. Thus for every growth rate r there is a unique constant-parameter BMVR that reproduces the renewal model's exponential solution exactly. What (87) cannot do is reproduce transients: a single pair $(p, d_{\mathcal{I}})$ matches one growth rate, and a transient samples a continuum of them (Figure 3 shows the exponential case; the overlays of §12.4 show everything else).

The two ends of the map $r \mapsto (p_{\text{eff}}, d_{\mathcal{I}, \text{eff}})$ are explicit. Using $\int_0^\infty \delta K = V_\infty$ (from $V' = \delta K$) and $\int_0^\infty \widehat{I} = \mathbb{E}(T_{\text{prod}})$ (Proposition 6.2),

$$r = 0 : \quad p_{\text{eff}}(0) = \frac{V_\infty}{\mathbb{E}(T_{\text{prod}})} = \frac{\beta V_\infty}{\log(a/(a-1))}, \quad d_{\mathcal{I}, \text{eff}}(0) = \frac{\beta}{\log(a/(a-1))}; \quad (89)$$

the $r = 0$ value of p_{eff} coincides with naive moment-matching — lifetime output divided by lifetime — a pleasant consistency check, and the matching used in the overlays of §12.4. At the other end,

$$r \rightarrow \infty : \quad p_{\text{eff}}(r) \rightarrow \delta, \quad d_{\mathcal{I}, \text{eff}}(r) \rightarrow \mu + \delta, \quad (90)$$

the *young-cell* limit: in a fast-growing epidemic almost every infected cell was infected moments ago, holds one particle, bursts at rate δ , and releases one particle when it does. Between the two ends, p_{eff} decreases monotonically in r across every parameter set we examined (Figure 10): fast epidemics are dominated by young cells and release slowly per cell; slow or stationary ones are dominated by old cells and release quickly. At $(\beta, \mu, \delta) = (1, 0, 0.1)$: $p_{\text{eff}}(0) \approx 4.59$ against $p_{\text{eff}}(\infty) = 0.1$ — the same pathogen, two orders of magnitude apart, depending only on how fast the epidemic around it is growing.

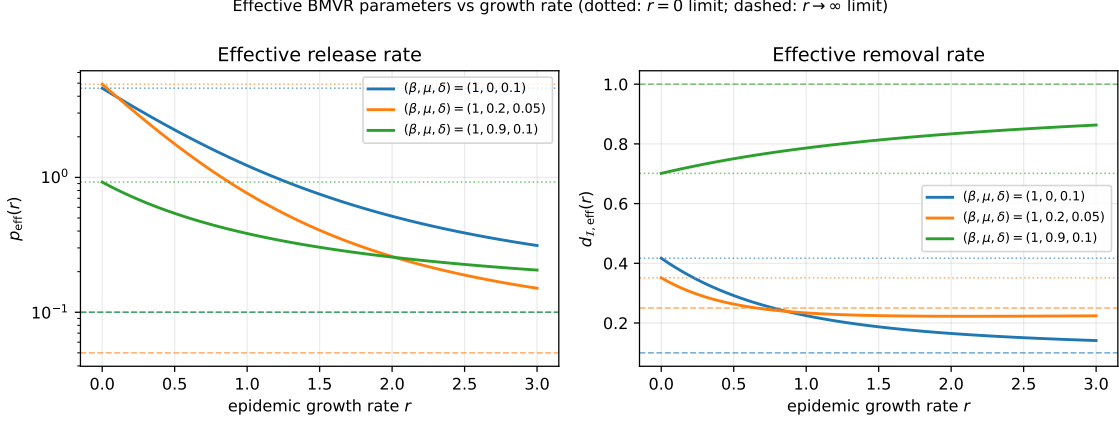


Figure 10: The effective release and removal rates (87) as functions of the epidemic growth rate r , for three parameter sets. Dotted and dashed lines mark the limits (89) and (90). p is not a constant of the pathogen; it is a function of the epidemic’s phase.

The closed hypergeometric forms of $\tilde{I}$ and $\delta\tilde{K}$ are given in Appendix C.1; the curves above are computed by direct quadrature.

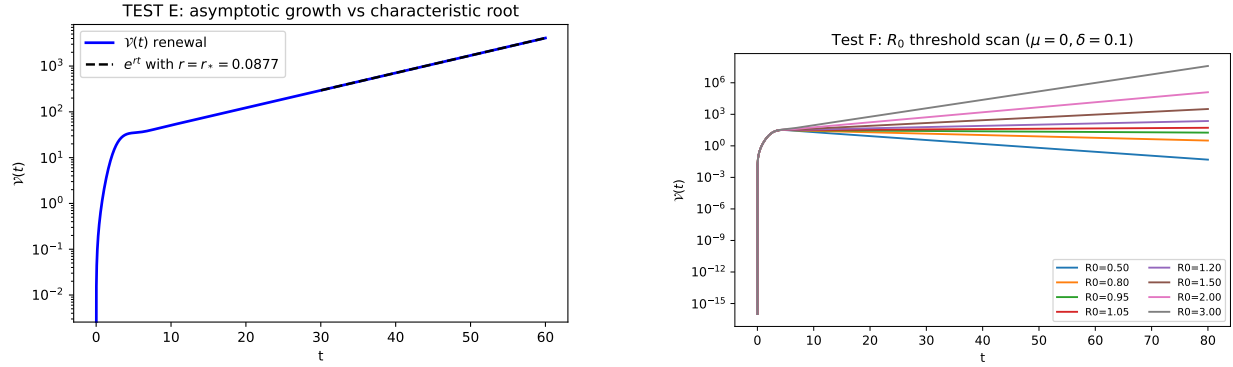
13.2 The characteristic equation and the invariance of R_0

Setting $r = 0$ in (88) gives the threshold condition $\gamma T \delta\tilde{K}(0) = c$, i.e. $\gamma T V_\infty = c$, so the basic reproduction number of the renewal model is

$$R_0^{\text{ODE}} = \frac{\gamma T V_\infty}{c}. \quad (91)$$

Theorem 13.1 (R_0 is unchanged by bursting). *At fixed total per-cell output V_∞ , replacing constant budding by bursting leaves R_0 exactly unchanged. Bursting alters the generation-time distribution — hence the growth rate r , hence the extinction probability, hence the stochastic early phase — but not the threshold itself.*

This is the classical R_0 -versus- r distinction in a setting where both sides are computable in closed form, and it connects the renewal model directly to the “perilous beginning” theme of Chapter 3 and to the flooding question of §14. Two cautions accompany it. First, the numerical checks of Figures 11a and 11b: the late growth rate of (81)–(82) agrees with the characteristic root r_* of (88) to $\sim 10^{-7}$ across five scenarios, and trajectories decay for $R_0 < 1$ and grow for $R_0 > 1$, the transition resolved already at $R_0 = 1.05$. Second, a notational care: (91) is the *continuum* reproduction number, in which one virion contributes mean $\gamma T/c$ new infections. A discrete branching process in which virions are absorbed on infection carries instead the mean $m = qV_\infty$, $q = \gamma T/(\gamma T + c)$; the two agree when $\gamma T \ll c$, and it is m that governs the branching calculations of §14. We write R_0 carefully in each context.



(a) Late growth = characteristic root.

(b) Sharp threshold at $R_0 = 1$.

Figure 11: Two structural checks of the renewal system: the fitted late growth rate of $\mathcal{V}$ against the root of (88) (five scenarios, relative error $\sim 10^{-6}$ – 10^{-7}), and the R_0 threshold scan (eight values through $R_0 = 1$).

13.3 Identifiability: a negative result

How much of the intracellular process can a population-level fit recover? At any fixed growth rate r , the map

$$(\beta, \mu, \delta) \mapsto (p_{\text{eff}}(r), d_{\mathcal{I}, \text{eff}}(r)) \quad (92)$$

sends three rates to two numbers, and is therefore not invertible. This closes, negatively, the ambition of the original Chapter 4 attempt, which hoped that a good estimate of p would grant access to the rates constituting the quasi-stationary mean: it would not, and neither would $(p, d_{\mathcal{I}})$ together. A third observable is required. In order of experimental accessibility, the natural candidates are:

1. the burst-size dispersion — the geometric ratio $1/a$ fixes one combination of the rates directly (§7.2);
2. the fraction of cells that clear without bursting, which is b ;
3. the shape — not merely the mean — of the release-time density $\varphi = \delta J$ (§7.1).

Any one of these, together with $(p, d_{\mathcal{I}})$, over-determines (β, μ, δ) and converts the fit into a consistency test of the whole intracellular model. The negative result is worth stating plainly: it is not a limitation of a fitting procedure but of the information content of constant-parameter BMVR itself.

13.4 Prediction for fitting practice

Since p_{eff} depends on r and decreases with it (Figure 10), BMVR fits to acute-phase data — fast growth, young cells — and to set-point data — slow or stationary — should yield systematically different estimates of p , in a direction and by an amount the theory specifies. Equivalently: the eclipse-like delay that within-host models usually insert by hand emerges here from the intracellular dynamics with no extra parameter at all — the renewal system’s early lag, visible in Figures 5 and 9, *is* the eclipse.

14 The flooding advantage and the growth-rate trade-off

Does bursting help a pathogen get established? The question belongs to the stochastic early phase, where censuses are small and extinction is a real threat, and it can now be answered exactly: both the bursting and the budding version of the early infection are branching processes whose offspring distributions are available in closed form.

14.1 Offspring of one infected cell

In the branching approximation each released particle independently either infects a target cell (rate γT) or is cleared (rate c), so it infects with probability

$$q = \frac{\gamma T}{\gamma T + c}. \quad (93)$$

The offspring of one infected cell is the burst size $\mathcal{K}$, thinned by this success probability. The (unconditional) burst-size probability generating function is, by Theorem 7.1, $G_{\mathcal{K}}(z) = b + (\delta/\beta) z/(a - z)$ — including the mass b of cells that never burst — and thinning replaces z by $y := 1 - q + qz$:

$$G_{\text{off}}(z) = b + \frac{\delta}{\beta} \frac{y}{a - y}, \quad y = 1 - q + qz. \quad (94)$$

One checks $G_{\text{off}}(1) = b + (\delta/\beta)/(a - 1) = b + B = 1$. The mean and variance of the offspring number are

$$m = G'_{\text{off}}(1) = q V_{\infty}, \quad \text{Var}_{\text{burst}} = \frac{2q^2 V_{\infty}}{a - 1} + q V_{\infty} - q^2 V_{\infty}^2, \quad (95)$$

the variance following from $\text{Var}(\mathcal{K})$ via the limit of (13) and the standard composition formula for random sums. When $\gamma T \ll c$, $q \approx \gamma T/c$ and $m \approx R_0^{\text{ODE}}$ of (91).

14.2 Extinction probability in closed form

The extinction probability of a branching process is the smallest root of $G_{\text{off}}(z) = z$ in $[0, 1]$. With $y = 1 - q + qz$, the fixed-point equation is quadratic, $z = 1$ is always a root, and the other root gives, for $m > 1$,

$$z_{\text{ext}}^{\text{burst}} = \frac{a(1 - q + qb) - 1 + q}{q} = \frac{a - 1}{q} + 1 - a(1 - b). \quad (96)$$

14.3 Comparison with matched-mean budding: the main theorem

The comparator is the classical picture: a cell with $\text{Exp}(d_{\mathcal{I}})$ lifetime releasing at constant rate p . The number it releases is a Poisson process stopped at an independent exponential time, hence geometric on $\{0, 1, 2, \dots\}$ with mean $p/d_{\mathcal{I}}$; matched to the bursting cell's total output, $p/d_{\mathcal{I}} = V_{\infty}$. Thinning a geometric law leaves a geometric law, so the budding offspring number is geometric with the same mean $m = qV_{\infty}$, and its extinction probability is

$$z_{\text{ext}}^{\text{bud}} = \frac{1}{m} \quad (m > 1). \quad (97)$$

Theorem 14.1 (The flooding advantage). *Write $L := a(1 - b) = a - \mu/\beta$, so that $V_{\infty} = L/(a - 1)$. For $m > 1$,*

$$z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L - 1) \left(\frac{1}{m} - 1 \right). \quad (98)$$

Hence bursting strictly lowers the establishment extinction probability, relative to matched-mean budding, if and only if $L > 1$, i.e.

$$\delta(\beta + \mu) > \beta\mu \iff \boxed{\frac{1}{\delta} < \frac{1}{\beta} + \frac{1}{\mu}} \quad (99)$$

(with the convention that the condition holds for all $\delta > 0$ when $\mu = 0$). When $L = 1$ the two extinction probabilities are identical for every q .

Proof sketch. The identity (98) is algebra: substitute (96), $m = qV_\infty = qL/(a - 1)$ and simplify. For the criterion, $L > 1$ is $a(1 - b) > 1$, i.e. $a > 1 + \mu/\beta$; inserting the definition of a and squaring (both sides are positive, or the right-hand side is not) yields $(\beta + \mu + \delta)^2 - 4\beta\mu > 4\beta^2(1 + \mu/\beta - \eta)^2$, which reduces to $\delta(\beta + \mu) > \beta\mu$. $\square$

Numerically, the identity (98) is exact in every case we checked:

(β, μ, δ)	q	$z_{\text{ext}}^{\text{burst}}$	$z_{\text{ext}}^{\text{bud}}$	verdict
(1, 0, 0.1)	0.2	0.400	0.455	bursting wins ($L = 1.10$)
(1, 0.9, 0.1)	0.9	0.935	0.844	budding wins ($L = 0.42$)
(1, 0.5, 1/3)	0.6	0.833	0.833	tie ($L = 1$)
(1, 0.5, 1/3)	0.9	0.556	0.556	tie ($L = 1$)

For $m \leq 1$ both processes die out almost surely and the comparison is empty; (98) changes sign there, an artefact of both formulae running outside their domain $[0, 1)$, not a reversal of the biology.

The interpretation is that the flooding advantage is real but bought against a risk. Bursting concentrates output into one batch — lowering the offspring variance relative to the geometric output of a budding cell — but it carries the all-or-nothing hazard that the intracellular population dies out first, with probability b , and the cell releases nothing. The condition says the advantage survives exactly when the burst timescale $1/\delta$ is shorter than the harmonic-type combination $1/(1/\beta + 1/\mu)$ of the birth and death timescales. When $\mu = 0$ the hazard of internal extinction vanishes and bursting always wins.

14.4 The boundary $L = 1$: a coincidence of laws

The boundary case is more than a tie of extinction probabilities.

Proposition 14.2. *The following are equivalent:*

$$L = 1 \iff a = 1 + \frac{\mu}{\beta} \iff \delta(\beta + \mu) = \beta\mu \iff \frac{1}{\delta} = \frac{1}{\beta} + \frac{1}{\mu}. \quad (100)$$

When they hold, the bursting offspring law (94) is exactly geometric with mean m — not merely equal in extinction probability to the budding comparator. In particular the offspring variances coincide, and

$$\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}} = \frac{2q^2V_\infty(L - 1)}{a - 1}, \quad (101)$$

so that at matched mean m the bursting offspring variance is smaller than the budding one when $L > 1$, equal when $L = 1$, and larger when $L < 1$.

Proof. $L = 1$ is $a(1 - b) = 1$, i.e. $a - 1 = ab = \mu/\beta$, the chain of equivalences above. Under it, $b = \delta/\beta$ (by (3)), and (94) collapses:

$$G_{\text{off}}(z) = b \left(1 + \frac{y}{a - y} \right) = \frac{ab}{a - y} = \frac{a - 1}{a - y} = \frac{(a - 1)/a}{1 - y/a},$$

using $ab = a - 1$. With $y = 1 - q + qz$,

$$1 - \frac{y}{a} = \frac{a - 1 + q}{a} \left(1 - \frac{q}{a - 1 + q} z \right),$$

so

$$G_{\text{off}}(z) = \frac{1 - r}{1 - rz}, \quad r = \frac{q}{a - 1 + q} = \frac{m}{1 + m}, \quad m = \frac{q}{a - 1} = qV_{\infty},$$

which is the probability generating function of a geometric law on $\{0, 1, 2, \dots\}$ with mean m — the same law as the matched budding comparator. The variance identity (101) is the difference between the geometric variance $m(1 + m)$ and (95), with $V_{\infty} = L/(a - 1)$. $\square$

At the boundary, then, bursting and budding are the same branching process in different clothes. Away from it, (101) quantifies the direction in which they differ.

14.5 The deterministic growth-rate trade-off

Establishment is only half of the story; the other half is how fast an established infection grows. Match the two models a second time, now at level R_0 and at mean productive lifetime: take the budding cell with $d_{\mathcal{I}} = 1/\mathbb{E}(T_{\text{prod}})$ and $p = V_{\infty}d_{\mathcal{I}}$ — which is exactly $p_{\text{eff}}(0)$, so this is the fairest possible classical match — and choose γT (or c) so that both models share $R_0 = 2$. Then solve for the deterministic growth rates: r_{bud} from the quadratic $(r + d_{\mathcal{I}})(r + c) = \gamma T p$, and r_{burst} from the characteristic equation (88). In every regime we checked — $L > 1$, $L = 1$, and $L < 1$ — the ordering is the same:

$$r_{\text{bud}} > r_{\text{burst}}. \tag{102}$$

With $c = 1$ and $R_0 = 2$:

(β, μ, δ)	L	$d_{\mathcal{I}}$	r_{bud}	r_{burst}
(1, 0, 0.1)	1.10	0.417	0.250	0.180
(1, 0.5, 1/3)	1.00	0.910	0.395	0.294
(1, 0.9, 0.1)	0.42	0.701	0.343	0.198

The mechanism is generation time: the budding cell starts releasing immediately, while the bursting cell must first accumulate load, so budding's generation times are shorter and its exponential invasion is faster — the same delay that appeared as the eclipse-like lag of Figures 5 and 9.

The two comparisons together give the biological headline. At fixed total output per cell, bursting can lower the probability of early extinction (when $L > 1$) while simultaneously slowing deterministic invasion speed: R_0 stays fixed, and z_{ext} and r move in opposite directions. A pathogen tuning its burst rate δ faces a genuine trade-off. For $\mu > 0$, budding beats bursting on extinction until δ crosses the threshold $\delta_* = \beta\mu/(\beta + \mu)$ — the $L = 1$ boundary — beyond which bursting takes over; for $\mu = 0$ bursting wins on extinction at every δ , at the continuing cost of a slower r .

Remark 14.3 (Scope of the comparison). Both results compare branching structures at fixed offspring mean. They say nothing about nonlinear effects — immune saturation, local depletion of targets, Allee-type cooperation — where other advantages of flooding may live; any full account should name and separate the two mechanisms. The evolutionary framing, and where this result sits relative to the bursting–budding spectrum of Chapter 6, is taken up there.

15 The spectrum of release models

Bursting and constant budding are the two ends of a spectrum, and the renewal construction of §12 is the bridge between them: *any* intracellular model that supplies a survival kernel $S(a)$ and a release kernel $g(a)$ induces the same population skeleton. This section lays out the spectrum. First the skeleton and the classical models built on it (§15.1–§15.2); then reset catastrophe, which interpolates by letting one cell release many times (§15.3); then self-exciting release (§15.4); and finally partial release, which interpolates in *how much* leaves the cell at an event (§15.5). The evolutionary comparison between the ends of the spectrum was made in §14; its ecological context belongs to Chapter 6. The whole arrangement is sketched in Figure 12.

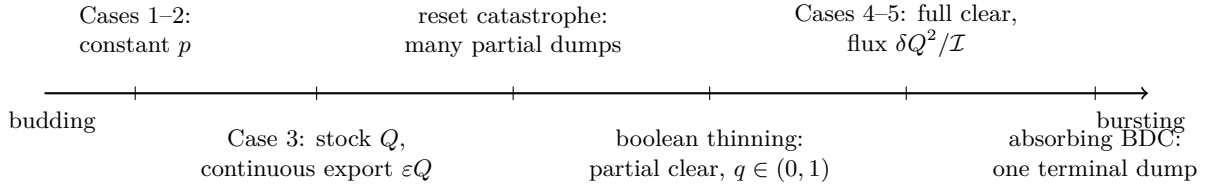


Figure 12: The budding–bursting spectrum. Moving right, release becomes more concentrated in time and more load-dependent: from constant budding (Cases 1–2) through stock-and-export (Case 3), recurrent partial dumps (reset catastrophe), a tunable fraction q per event (boolean thinning), load-proportional full clear (Cases 4–5), to the single terminal dump of the absorbing BDC of this chapter.

15.1 The universal renewal skeleton

With incidence $i(t) = \gamma T \mathcal{V}(t)$ (or a nonlinear incidence $f(\mathcal{V}, T)$), any choice of kernels gives

$$\mathcal{I}(t) = \mathcal{I}_0 S(t) + (i * S)(t), \quad \frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g(t) + (i * g)(t) - c \mathcal{V}(t), \quad (103)$$

with effective parameters $p_{\text{eff}}(r) = \tilde{g}(r)/\tilde{S}(r)$ and, for linear incidence, $R_0 = (\gamma T/c) \int_0^\infty g$. The models of this section differ only in their (S, g) :

Model	$S(a)$	$g(a)$
Classical BMVR	$e^{-d_{\mathcal{I}} a}$	$p e^{-d_{\mathcal{I}} a}$
Absorbing BDC (this chapter)	$\hat{I}(a)$	$\delta K(a)$
MOI- k BDC (§9)	$I(a)^k - D(a)^k$	$\delta K_k(a)$
Reset catastrophe (§15.3)	$e^{-d_{\mathcal{I}} a}$	$\delta e^{-d_{\mathcal{I}} a} \mathbb{E}(X_a^2)$
HIV multi-stage (§16)	stage survival	$p \times$ productive occupancy
Two-type / GATE (Ch. 5)	two-type $\hat{I}$	two-type second-moment flux

The first row is the check of Figure 3: exponential kernels reduce (103) exactly to classical BMVR.

15.2 Five modular ODE cases

When a finite-dimensional ODE description is wanted — for fitting, or for analytic tractability — the same ideas assemble into a ladder of five cases. Target cells T are held fixed throughout; $\mathcal{I}$ denotes productively infected cells, $\mathcal{V}$ free particles, E eclipse cells, and Q the total intracellular stock summed over infected cells. In all five cases, production into the stock (once productive) is *mean-linear*, immigration-type, at rate ν per cell — the choice appropriate to template-limited intracellular accumulation (§16.4); what distinguishes the cases is the release law and the presence or absence of an eclipse class. In this subsection α denotes the eclipse conversion rate; it is not the root a .

15.2.1 Case 1 — Classic BMVR

An infected cell produces particles that are released immediately at constant rate p (budding). No eclipse, no stored pool; infected cells are removed at rate $d_{\mathcal{I}}$:

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad R_0 = \frac{\gamma T p}{c d_{\mathcal{I}}}. \quad (104)$$

Production and export are collapsed into the single phenomenological parameter p . This is the model the rest of the chapter is replacing.

15.2.2 Case 2 — Eclipse + classic release

New infections enter an eclipse class E — infected but not yet producing — and convert to productive cells at rate α (mean eclipse length $1/\alpha$), with optional eclipse death d_E :

$$\frac{dE}{dt} = \gamma T \mathcal{V} - (\alpha + d_E) E, \quad \frac{d\mathcal{I}}{dt} = \alpha E - d_{\mathcal{I}} \mathcal{I}, \quad \frac{d\mathcal{V}}{dt} = p \mathcal{I} - c \mathcal{V}, \quad (105)$$

$$R_0 = \frac{\gamma T}{c} \cdot \frac{\alpha}{\alpha + d_E} \cdot \frac{p}{d_{\mathcal{I}}}. \quad (106)$$

A fraction $\alpha/(\alpha + d_E)$ of eclipse cells become productive; each then yields mean $p/d_{\mathcal{I}}$ particles. Setting $\alpha \rightarrow \infty$ with $d_E = 0$ recovers Case 1. A more realistic eclipse-time distribution is an Erlang chain $E_1 \rightarrow \dots \rightarrow E_n \rightarrow \mathcal{I}$, each step at rate α ; the $\mathcal{V}$ equation is unchanged. This is the standard delay construction of the within-host literature — and, as §16 will emphasise, it is inserted by hand there, whereas in the two-type BDC it emerges mechanically (the GATE observation of Chapter 5).

15.2.3 Case 3 — Intracellular stock + continuous export

Productively infected cells accumulate stock at rate ν per cell; each intracellular unit is exported at per-capita rate ε ; cell death removes the cell and its share of stock:

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \varepsilon Q - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \varepsilon Q - c \mathcal{V}. \quad (107)$$

The mean load per productive cell is $m = Q/\mathcal{I}$. In the quasi-steady stock regime ($\frac{dQ}{dt} \approx 0$),

$$Q_* = \frac{\nu}{\varepsilon + d_{\mathcal{I}}} \mathcal{I}, \quad p_{\text{eff}} := \frac{\varepsilon Q_*}{\mathcal{I}} = \frac{\varepsilon \nu}{\varepsilon + d_{\mathcal{I}}}, \quad (108)$$

and Case 3 collapses to Case 1 with $p = p_{\text{eff}}$: *classical BMVR is the fast-export limit of stock-and-export dynamics*. When export is not fast, $\mathcal{V}$ lags and depends on the history of Q . With eclipse stacked in front (incidence into E , conversion at α , only $\mathcal{I}$ producing), the Q and $\mathcal{V}$ equations are unchanged and eclipse cells contribute nothing until conversion. The reproduction number is

$$R_0 = \frac{\gamma T \nu \varepsilon}{c d_{\mathcal{I}}(\varepsilon + d_{\mathcal{I}})} \quad (\text{no eclipse}), \quad (109)$$

which is Case 1's R_0 with $p = \varepsilon \nu / (\varepsilon + d_{\mathcal{I}})$.

15.2.4 Case 4 — Intracellular stock + proportional full clear

The same accumulation, but release now comes in load-proportional dumps: in a cell with load x , dumps occur at rate δx and the whole pool is transferred (the cell survives the dump; death remains the separate rate $d_{\mathcal{I}}$). Under the homogeneous-load closure $x \approx m = Q/\mathcal{I}$, the population dump flux is $\delta \mathbb{E}(\sum_i X_i^2) \approx \delta \mathcal{I} m^2 = \delta Q^2/\mathcal{I}$, giving

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \delta \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \delta \frac{Q^2}{\mathcal{I}} - c \mathcal{V}, \quad (110)$$

with the convention that the nonlinear terms vanish when $\mathcal{I} = 0$. The dump term is simultaneously export into $\mathcal{V}$ and clearance of Q ; it is quadratic in total load at fixed $\mathcal{I}$, i.e. superlinear in mean load — the mean-field echo of the superlinearity of g_k in §9. The mature quasi-steady balance, $\nu = \delta m^2 + d_{\mathcal{I}} m$, gives

$$m = \frac{-d_{\mathcal{I}} + \sqrt{d_{\mathcal{I}}^2 + 4\delta\nu}}{2\delta}, \quad p_{\text{eff}} = \delta m^2, \quad (111)$$

which plays the role of p only in that steady regime: transients and early infection need the full $(Q, \mathcal{V})$ system. For invasion analysis near $\mathcal{I} = 0$, the map $Q \mapsto Q^2/\mathcal{I}$ is singular; one should linearise the underlying single-cell process instead, taking $R_0 = \gamma T V_{\infty}/c$ as in the renewal framework.

15.2.5 Case 5 — Eclipse + stock + proportional full clear

Case 2's eclipse block in front of Case 4: incidence creates eclipse cells; only productive cells accumulate stock and dump:

$$\frac{dE}{dt} = \gamma T \mathcal{V} - (\alpha + d_E) E, \quad \frac{d\mathcal{I}}{dt} = \alpha E - d_{\mathcal{I}} \mathcal{I}, \quad (112)$$

$$\frac{dQ}{dt} = \nu \mathcal{I} - \delta \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \delta \frac{Q^2}{\mathcal{I}} - c \mathcal{V}. \quad (113)$$

Eclipse adds a further staged lag — E rises first, then $\mathcal{I}$ and Q , then $\mathcal{V}$ — which Case 4 cannot produce; the quasi-steady relations for the mean load among productive cells are unchanged. In the single-cell sense,

$$R_0 = \frac{\gamma T}{c} \cdot \frac{\alpha}{\alpha + d_E} \cdot V_{\infty}. \quad (114)$$

15.2.6 Summary of the five cases

Case	Intracellular story	Free-particle source term
1 Classic	none (instant release)	$p\mathcal{I}$
2 Eclipse + classic	delay $E \rightarrow \mathcal{I}$, then as 1	$p\mathcal{I}$
3 Stock + continuous export	ν into Q , export εQ	εQ
4 Stock + full clear	ν into Q , dump $\propto Q^2/\mathcal{I}$	$\delta Q^2/\mathcal{I}$
5 Eclipse + stock + full clear	$E \rightarrow \mathcal{I}$, then as 4	$\delta Q^2/\mathcal{I}$

What is fixed across Cases 3–5 is the mean-linear production; what is chosen is the release law and the presence of eclipse, and Case 5 is exactly Case 4 composed with Case 2’s eclipse block. Each case also induces kernels (S, g) , so the convolution form (103) remains available whenever the ODE stage structure is replaced by age structure — which is precisely the absorbing BDC at the bursting end of the ladder.

15.3 Reset catastrophe

The absorbing BDC releases at most once per cell. An intermediate worth isolating lets a cell release *many* times: while the cell is alive, the load undergoes birth βn , death μn , optional immigration ν , and a *dump that resets*: at rate δn the whole pool n is released and the load returns to 0, after which it may re-accumulate. The cell itself dies at a separate rate $d_{\mathcal{I}}$, independent of the load. Biologically, read the load as a local assembled pool rather than the whole-cell contents: full dumping is one mathematical extreme of batch release, budding is many small exports, and this process is a tractable multi-pulse caricature between them.

Because dumps do not absorb, the kernels change but the skeleton does not:

$$S(a) = e^{-d_{\mathcal{I}}a}, \quad g(a) = \delta e^{-d_{\mathcal{I}}a} \mathbb{E}(X_a^2), \quad (115)$$

where $\mathbb{E}(X_a^2)$ is computed for the non-absorbing reset process. Linear rates keep the moments of that process tractable (closed ODEs under the generator), so g is explicit once they are solved; structurally, $g \propto \mathbb{E}(X^2)$ exactly as for the absorbing BDC. The renewal BMVR is identical in form to (81)–(82), with

$$p_{\text{eff}}(r) = (r + d_{\mathcal{I}}) \tilde{g}(r), \quad d_{\mathcal{I},\text{eff}} \equiv d_{\mathcal{I}}, \quad (116)$$

and $R_0 = (\gamma T/c) \int_0^\infty g$ — finite only for $d_{\mathcal{I}} > 0$, since otherwise recurrent dumps could continue forever. If the reset process admits a stationary law π and $d_{\mathcal{I}}$ is small, the mature-cell limit is

$$p_\infty = \lim_{a \rightarrow \infty} \frac{g(a)}{S(a)} = \delta \mathbb{E}_{(\cdot)} \pi[X^2] : \quad (117)$$

a long-lived infected cell behaves as classical constant- p BMVR with that p , while transients (first dumps, young cells) still need the full kernel. An eclipse is added exactly as before: set $g(a) = 0$ for $a < \tau_e$ and keep $S(a) = e^{-d_{\mathcal{I}}a}$.

	Absorbing BDC	Reset catastrophe
After dump	process ends in R	$X \rightarrow 0$, continues
Dumps per cell	≤ 1	many, until cell death
Survival kernel	$\hat{I}$	$e^{-d_I a}$
Release kernel	$\delta K_{\text{BDC}}(a)$	$\delta e^{-d_I a} \mathbb{E}(X_a^2)$
Population equations	renewal BMVR	same form
Mature limit	not constant budding	$p \rightarrow \delta \mathbb{E}(\cdot) \pi[X^2]$

15.4 Self-exciting release

A further biological possibility is that release is partially *self-exciting*: one event makes further release more likely for a short time — cooperative nucleation, local membrane platforms, ESCRT machinery already recruited at a site. The possibility must be balanced against depletion of stock and finite machinery: pure open-ended self-excitation is implausible, and a hybrid of excitation, refill and depletion is the reasonable picture. Two levels of model are available, as throughout: a renewal BMVR exists whenever the single-cell model defines (S, g) ; a finite ODE BMVR needs the excitation carried by a low-dimensional Markov state.

The menu of candidates:

#	Model	Excitation carried by	ODE BMVR?	Renewal?
1	Classical Hawkes	event history only	poor	yes (numerical g)
2	Hawkes + stock X	events + depletion/refill	via Model 10	yes
3	Marked Hawkes	size-weighted events	approximate	yes (numerical)
4	ETAS-style	size + time kernel	approximate	yes (numerical)
5	Cox + latent Z	latent excitability	if Z simple	yes
6	ON/OFF telegraph	sustained high phase	yes	yes
7	$\lambda \propto X^2$	current load only	yes (Cases 4–5)	yes
8	Nucleation biophysics	assembly cascade	hard	yes (numerical)
9	Age since last release	renewal excitation	yes (stages)	yes
10	Jumping rate $r(t)$	shot-noise intensity	yes (mean-field)	yes
11	Spatial Hawkes	local membrane	average first	after coarse-graining
12	Full clear (Cases 4–5)	instant total cascade	yes	yes

They nest: Model 2, with intensity $\lambda(t) = f(X_t) + \sum_{t_i < t} \phi(t - t_i)$, is the general event-history form — $\phi \equiv 0$ and $f(x) = \varepsilon x$ recovers continuous export (Case 3); $\phi \equiv 0$, $f(x) = \delta x^2$ with full clear recovers Model 7 and Cases 4–5; an exponential $\phi(t) = \eta e^{-\kappa t}$ is equivalent to Model 10; Model 6 is a coarse two-level version without a continuous intensity. The four focal models:

Model 7 — cooperative intensity, $\lambda(x) = \delta x^2$ (or $\delta x(x - 1)$). No memory of past events, only the current load: more material makes release disproportionately more likely. The single-cell process is production ν , optional degradation μX , and full-clear dumps at rate δX^2 ; the ODE form is exactly Cases 4–5 under the homogeneous closure, and the renewal form has $g(a) = \delta \mathbb{E}(X_a^2 \mathbf{1}\{\text{alive}\})$. Cooperativity without history; the default bulk “excitatory” model.

Model 6 — ON/OFF telegraph. The cell (or a membrane site) switches between OFF and ON at rates σ_{on} , σ_{off} ; while ON it releases at rate ρ (or εX). Sample paths are intermittent surges

without full clear. The simplest ODE version splits the infected class:

$$\frac{dI_{\text{OFF}}}{dt} = \gamma T \mathcal{V} + \sigma_{\text{off}} I_{\text{ON}} - (\sigma_{\text{on}} + d_{\mathcal{I}}) I_{\text{OFF}}, \quad (118)$$

$$\frac{dI_{\text{ON}}}{dt} = \sigma_{\text{on}} I_{\text{OFF}} - (\sigma_{\text{off}} + d_{\mathcal{I}}) I_{\text{ON}}, \quad \frac{d\mathcal{V}}{dt} = \rho I_{\text{ON}} - c \mathcal{V}, \quad (119)$$

and quasi-steady switching yields effective budding at the duty cycle, $p_{\text{eff}} = \rho \sigma_{\text{on}} / (\sigma_{\text{on}} + \sigma_{\text{off}})$ — a transparent limit back to constant p . Clustered release with writable ODEs.

Model 10 — jumping continuous release rate. Keep a continuous intensity $r(t) \geq 0$ that jumps up by η at each release event and decays to baseline r_0 at rate κ : $dr = -\kappa(r - r_0)dt + \eta dN_t$, with flux $r(t)$ or $r(t)X_t$. This is the Markovian (shot-noise) form of Hawkes with exponential kernel. Tracking $\mathcal{I}$, total load Q and total excitation $R = \sum r_i$ ($\bar{r} = R/\mathcal{I}$), a mean-field closure gives ODEs for $(\mathcal{I}, Q, R, \mathcal{V})$ with flux $\approx \bar{r}Q$; true fading-memory self-excitation with finite-dimensional state, at the cost of closures and three extra parameters.

Model 2 — Hawkes with stock. The literal version: $\lambda(t) = f(X_t) + \sum_{t_i < t} \phi(t - t_i)$; on an event X drops by the released amount; between events X refills at rate ν ; the cell dies at rate $d_{\mathcal{I}}$. The renewal form is always well-defined ($g(a) = \mathbb{E}(\lambda(a) \mid \text{alive})$), typically numerical; an ODE form exists only after reduction to exponential ϕ , i.e. to Model 10. The right language for “Hawkes-like budding” in a paper; the right implementation is Model 10.

	7 $\lambda \propto X^2$	6 ON/OFF	10 jumping r	2 Hawkes+ X
Carrier	load only	discrete phase	continuous r	event history (+load)
Memory	Markov in X	Markov in phase	Markov in r	non-Markov unless exp. ϕ
Clusters	high $X \rightarrow$ fast dumps	long ON sojourn	high- r window	aftershocks
ODE BMVR	yes (Cases 4–5)	yes (split $\mathcal{I}$)	yes (mean-field R)	only via 10
Renewal	yes	yes	yes	yes
$\rightarrow$ classic p	QSS δm^2	duty cycle	QSS $\mathbb{E}(rX)$	as 10 if exp. ϕ

Biologically, cooperative assembly makes short clustered release a medium–high confidence hypothesis; a bare Hawkes process with event memory and no depletion is a low–medium confidence standalone cell model. The practical ranking: Model 7 for writable ODEs inside Cases 3–5; Model 6 for intermittent release with the simplest new ODEs; Model 10 for fading memory with ODEs; Model 2 as the conceptual definition.

15.5 Boolean and partial release

The last axis of the spectrum is *how much* leaves the cell when a release event fires. Cases 3–5 sit at two extremes — continuous export vs full clear — and a natural intermediate removes only a random part of the load: at load n , release B with $0 \leq B \leq n$, retain $n - B$. Two ingredients are free: when events fire (rate $\lambda(X)$): constant δ , linear δX , or cooperative δX^2), and the law of $B \mid X = n$. The options:

- A. **Boolean thinning (recommended default):** each unit is released independently with probability q , $B \mid X = n \sim \text{Binomial}(n, q)$; $q \rightarrow 0$ tiny fractional clear, $q = 1$ full clear. Generating functions stay friendly.
- B. Fixed fraction $B = \text{round}(fn)$: same mean as A with $q = f$, less stochasticity.
- C. Random fraction ($U \sim \text{Beta}$ or uniform thinning): more variable bursts than B.
- D. Fixed capped batch $B = \min(k, n)$: $k = 1$ is unit budding in the frequent-event limit; large k approaches full clear.
- E. Size-biased release, $P(B = j \mid n) \propto j$: natural if one site releases a random assembled blob.
- F. Threshold-then-partial: events only for $X \geq x_*$, then A or B — accumulation with a trigger.
- G. Short high-export window: a triggered burst of high ε for a brief time — partial clear in expectation, related to Model 6.

The single-cell boolean process: produce at rate ν ; release events at rate $\lambda(X)$ with $B \sim \text{Binomial}(X, q)$ released and $X \rightarrow X - B$; cell death at $d_{\mathcal{I}}$. Its nested limits:

Choice	Recovers
$q = 1, \lambda = \delta X^2$	full clear, cooperative (Case 4 style)
$q = 1, \lambda = \delta X$	full clear, linear hazard
$0 < q < 1, \lambda = \delta X$	the main partial-clear intermediate
$B \equiv 1, \lambda = \varepsilon X$	unit budding (Case 3, discrete)

At the mean-field level with $\lambda = \delta X$, the population equations are Case 4 scaled by q :

$$\frac{d\mathcal{I}}{dt} = \gamma T \mathcal{V} - d_{\mathcal{I}} \mathcal{I}, \quad \frac{dQ}{dt} = \nu \mathcal{I} - \delta q \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q, \quad \frac{d\mathcal{V}}{dt} = \delta q \frac{Q^2}{\mathcal{I}} - c \mathcal{V}. \quad (120)$$

With a constant clock $\lambda = \delta$ instead, the mean export is $\delta q Q$ and the bulk ODEs collapse to Case 3 with $\varepsilon_{\text{eff}} = \delta q$: the pulses remain visible in single-cell trajectories and higher moments, but not in the first-moment ODE. Indeed, partial vs full release is invisible in mean ODEs whenever the event rate is constant; it becomes visible in bulk dynamics when the clock is load-dependent, and in single-cell release-count distributions always.

The recommendation: boolean thinning with a load-dependent clock ($\lambda = \delta X$ or δX^2) as the primary intermediate — one transparent parameter q between mild partial clear and full dumping, mechanistic, nested in Cases 3–4 at the mean-field level, easy to simulate and to write BMVR ODEs for; fixed fraction as a secondary when less burst-size noise is wanted. The flooding boundary as a function of q — where on this continuum the advantage of §14 survives — is open.

16 Contrast with HIV: what not to claim

The core results of this chapter are written for pathogens whose intracellular life ends in load-proportional catastrophe — lyse and dump. The motivating applications (*Y. pestis*, *F. tularensis*, lytic phage) are of that kind. HIV is not: it buds from the plasma membrane, and an infected cell need not rupture and empty itself in one event. Yet single-cell and limiting-dilution data show something that looks surge-like — long quiet periods, then a concentrated bout of release

— which is exactly the phenomenon the BDC explains. The honest position is neither to force HIV into the pure BDC nor to retreat to constant- p BMVR, but to build an eclipse-structured, short-productive-pulse model on the same renewal skeleton. This section records what the best data support, gives candidate equations, and draws the boundary of the chapter’s claims.

16.1 What Hataye et al. (2019) show

Hataye and co-authors [11] coupled limiting-dilution latency-reactivation cultures to stochastic models, and their findings — summarised here at the level that bears on equation choice — fall in two groups.

On initial release (viral inhibition cultures; no de novo infection).

1. The timing and magnitude of first HIV RNA release from reactivated latently infected CD4 T cells are highly variable: total yield, delay to first detection, and detection duration all fluctuate.
2. Mean release is delayed: first detection most often around day 3, with a sudden bulk of production around days 3–5 — not a flat continuous drip from day 0.
3. A simple productive-cell model (release at constant rate p , death at rate δ_I , total yield geometric with mean p/δ_I) is under-dispersed relative to the data: the Fano factor is too small.
4. A single exponential eclipse $E \rightarrow I$ fits the delay poorly (its mode sits at day 1). A multi-stage eclipse — an Erlang-like chain of n compartments, with $n \approx 5$ favoured — fixes the mean timing.
5. Multi-stage eclipse alone still under-disperses, and transcriptional ON/OFF toggling does not fix the dispersion under realistic CD4 division/death rates ($\rho_{\text{div}} \approx \mu_E \approx 0.5/\text{day}$ from CFSE and population stability data).
6. Their best structure (“Model 3”) uses two distinct eclipse paths (high- and low-release potential), multi-stage delay, and division/death in eclipse. Then: many stimulated lineages release nothing detectable ($f_{\text{HIV}} \approx 0.41$ of single latently infected cells give detectable virus in their fit); most productive sojourns last on the order of a day; and multi-day “sustained” detection is largely proliferation in eclipse followed by staggered short $E \rightarrow I$ transitions of descendants — not one cell budding steadily for a week.

On establishment (outgrowth cultures; de novo infection allowed).

1. Release of replication-competent virus often fails to establish exponential spread.
2. Establishment probability as a function of the expected number of virus-releasing cells per well is sigmoid, not the concave curve of independent per-cell success — interpreted as an Allee effect, synergy at low census.
3. A critical threshold sits near $\Lambda \approx 2.3$ latently infected cells, corresponding on average to roughly 5100 detected HIV RNA copies of initial release (under their establishment definition of $\geq 2 \times 10^5$ copies).

4. For exactly one stimulated latently infected cell, their fitted dynamics predict establishment only on the order of 2% of the time — consistent with the known gap between intact provirus counts and quantitative viral outgrowth.

Supported for HIV	Not supported as literal HIV
Multi-stage eclipse (delay)	Pure one-type BDC terminal catastrophe
Short productive pulse then cell death	Memoryless constant- p BMVR from $t = 0$
High stochasticity; many silent lineages	Independent branching alone near threshold
Renewal / stage structure for free virus	QSD = geometric burst size as an HIV law
Allee / synergistic establishment	

The visual “surge” is real, but in their best model it is delay plus brief constant-rate production plus latent lineage dynamics — not a single lytic dump of intracellular load.

	Lytic BDC (this chapter)	HIV (Hataye-guided)
Release mechanism	catastrophe at rate δX	budding while productive
End of production	terminal; process absorbed	short productive life; cell dies
Intracellular load X	explicit birth–death state	not required as primary state
Quiet period	until first catastrophe	multi-stage eclipse (+ latency)
Sustained multi-day release	one large dump (or nothing)	often staggered short pulses
Establishment theory	independent offspring as baseline	Allee synergy near threshold
Natural hosts	<i>Y. pestis</i> , <i>F. tularensis</i> , phage	CD4 T-cell latency / rebound

16.2 Candidate HIV stage equations

A compact Hataye-aligned system, suitable for coupling to the renewal free-virus layer of Result 12.1 (Figure 13). Let L_0 be a stimulated latently infected cell (or the start of reactivation). Two eclipse pathways run in parallel: A , of high release potential and multi-stage, and B , low and brief:

$$L_0 \xrightarrow{\alpha_A} E_1^{(A)}, \quad L_0 \xrightarrow{\alpha_B} E^{(B)}, \quad (121)$$

$$E_j^{(A)} \xrightarrow{\alpha} E_{j+1}^{(A)} \quad (j = 1, \dots, n-1), \quad E_n^{(A)} \xrightarrow{\alpha} I^{(A)}, \quad E^{(B)} \xrightarrow{\alpha_B} I^{(B)}. \quad (122)$$

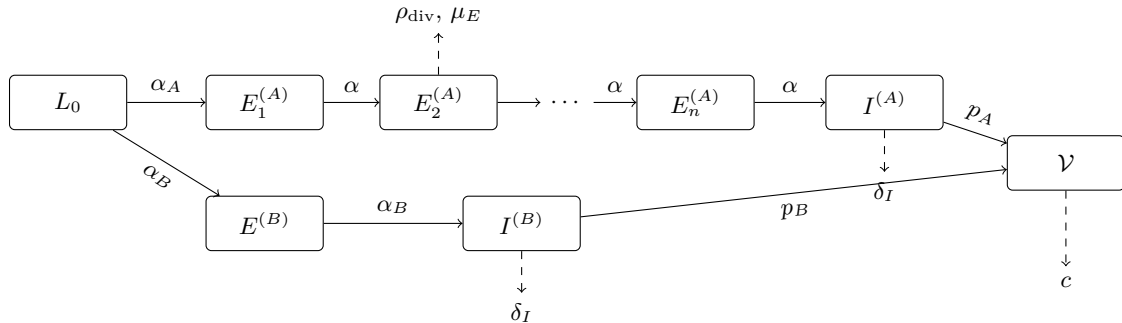


Figure 13: The Hataye-aligned stage structure of §16.2: two eclipse pathways from the stimulated latent class L_0 — a multi-stage path A of high release potential and a brief path B — each feeding a short productive class that releases at constant rate; division and death (dashed) act in eclipse, death δ_I in the productive classes, and clearance c on free virus.

In each eclipse compartment, cells may divide or die without releasing virus,

$$E \xrightarrow{\rho_{\text{div}}} E + E, \quad E \xrightarrow{\mu_E} \emptyset, \quad (123)$$

with $\rho_{\text{div}} \approx \mu_E \approx 0.5/\text{day}$ in the Hataye fits. Productive cells release at constant rate and die,

$$I^{(A)} \xrightarrow{p_A} I^{(A)} + \mathcal{V}, \quad I^{(A)} \xrightarrow{\delta_I} \emptyset, \quad I^{(B)} \xrightarrow{p_B} I^{(B)} + \mathcal{V}, \quad I^{(B)} \xrightarrow{\delta_I} \emptyset, \quad (124)$$

with $p_A \gg p_B$ (or p_B small and $I^{(B)}$ short-lived) to match the bimodal release distribution; free virus clears at rate c . The mean-field stage ODEs (large inoculum Λ , suppressing the dual path for display) are

$$\frac{dE_1}{dt} = \alpha_A L_0 \delta_0(t) + \rho_{\text{div}} E_1 - (\mu_E + \rho_{\text{div}} + \alpha) E_1, \quad (125)$$

$$\frac{dE_j}{dt} = \alpha E_{j-1} + \rho_{\text{div}} E_j - (\mu_E + \rho_{\text{div}} + \alpha) E_j, \quad j = 2, \dots, n, \quad (126)$$

$$\frac{dI^{(A)}}{dt} = \alpha E_n - \delta_I I^{(A)}, \quad \frac{d\mathcal{V}}{dt} = p_A I^{(A)} - c \mathcal{V} \quad (+ p_B I^{(B)}), \quad (127)$$

where the Dirac source stands for an initial cohort; for ongoing infection it is replaced by incidence, split between pathways with $\pi_A + \pi_B = 1$:

$$i(t) = f(\mathcal{V}(t), T), \quad \text{source into } E_1^{(A)} = \pi_A i(t), \quad \text{source into } E^{(B)} = \pi_B i(t). \quad (128)$$

Linear infection (the classical, large-population limit) takes $f(\mathcal{V}, T) = \gamma T \mathcal{V}$ and recovers a stage-structured BMVR. Near the establishment threshold, however, the Hataye data favour a sigmoid law over independent per-virion success; a minimal phenomenological upgrade is the Allee incidence

$$f(\mathcal{V}, T) = \gamma T \mathcal{V} \cdot \frac{\mathcal{V}}{K_A + \mathcal{V}}, \quad (129)$$

inefficient at very low $\mathcal{V}$ and saturating to mass action for $\mathcal{V} \gg K_A$, with K_A chosen so that cumulative initial release near $\sim 5 \times 10^3$ RNA copies sits at the inflection of establishment probability — the empirical threshold above. More mechanistic forms (cooperative entry, restriction-factor saturation, local multiplicity) can replace it; the point is that linearity in $\mathcal{V}$ alone is not faithful near establishment.

Coupled to free virus, the stage model induces renewal kernels of the usual kind. Let $g_{\text{HIV}}(a)$ be the expected release rate at age a after a single infection or reactivation, and $S_{\text{HIV}}(a)$ the probability of still being in a productive (or not-yet-cleared) lineage. Then exactly the skeleton (103) holds:

$$\mathcal{I}(t) = \mathcal{I}_0 S_{\text{HIV}}(t) + (i * S_{\text{HIV}})(t), \quad \frac{d\mathcal{V}}{dt} = \mathcal{I}_0 g_{\text{HIV}}(t) + (i * g_{\text{HIV}})(t) - c \mathcal{V}(t). \quad (130)$$

For the stage model, g_{HIV} is *not* δK : it is the density of productive sojourn times times p_A (and p_B), with the Erlang eclipse supplying the delay — $g_{\text{HIV}}(a) \approx 0$ for small a , then rising and falling: the mathematical expression of “nothing for a long time, then a lot”. The effective parameters of §13 carry over by the substitution $(\hat{I}, g) \mapsto (S_{\text{HIV}}, g_{\text{HIV}})$:

$$p_{\text{eff}}(r) = \frac{\tilde{g}_{\text{HIV}}(r)}{\tilde{S}_{\text{HIV}}(r)}, \quad R_0 = \frac{\gamma T}{c} \int_0^\infty g_{\text{HIV}}(a) da \quad (\text{linear incidence}), \quad (131)$$

while under the Allee form R_0 remains the large-population threshold and early establishment is governed in addition by the scale K_A .

16.3 What to keep, what not to claim

- **Keep** for HIV: the renewal free-virus coupling; age/stage kernels; the lesson that constant p is only an exponential-phase projection; stochastic early failure; eclipse as a real dynamical object (the GATE mechanism of Chapter 5, multi-stage here).
- **Do not claim** for HIV: terminal catastrophe at rate δX ; QSD = geometric burst size as an HIV law; one-shot cumulative release as the only single-cell trajectory; independent branching as the full establishment story near the Hataye threshold.
- **Flooding comparison:** still meaningful as a *linear* baseline — matched mean yield, compare offspring variance — but near $\sim 5 \times 10^3$ RNA copies synergy can dominate variance effects, and any HIV write-up should separate the two mechanisms.

16.4 HIV intracellular mean growth: linear vs exponential

One modelling decision underlies the stock-and-export cases of §15.2 and deserves its own statement: does the mean intracellular load grow exponentially (birth–death) or linearly (template-limited production)?

Regime	Mean law	Mechanism
Mean-exponential	$m' = r m$	each unit generates more units (branching / BDC)
Mean-linear	$m' \approx \nu - \mu m$	production set by fixed templates / machinery

Absorbing BDC and pure birth–death are exponential-mean; immigration and constant production are linear-mean. HIV does *not* amplify infectious units inside the cell by branching: after integration the DNA template count is typically one or a few proviruses, not an exponentially growing genome pool; late genomic RNA is made by host Pol II from that fixed template, not by cytoplasmic genome cascading; and Gag and the other structural products are translated from mRNA and do not self-replicate. Production is immigration-like: rate set by mRNA and ribosomes, minus degradation, packaging and export. Pure $m' = r m$ is a poor literal description of HIV intracellular load; it fits phage genome races, intracellular bacterial division, and some cytoplasmic RNA viruses far better. The main nonlinearity is Tat positive feedback: transcription rises from a low basal level towards a saturated high state. During the switch, product counts can rise faster than a pure linear ramp, because the *production rate* is increasing — not because each unit is a branching particle; and once fully ON, production is roughly constant about a high mean, so mean accumulation is linear or saturating. The late fully-ON rate is the ceiling, and the intermediate phase climbs towards it; in the mean it does not systematically exceed it (stochastic transcriptional bursts can spike above the late *mean* rate, but that is noise, not a hotter phase; and net accumulation can peak mid-course if export turns on late, but that is export balance, not Tat overshoot). Confidence: high that HIV is not pure exponential-mean birth–death of virion-like units; medium–high that the main productive window is closer to mean-linear than exponential; medium on the exact shape of $m(a)$, which depends on the molecular species counted and on the Tat phase. The recommended coarse regime is therefore

$$\text{eclipse} \longrightarrow \text{mean-linear (immigration-type) accumulation of } X, \quad (132)$$

with a fixed or multi-step delay so first activity is not peaked at $t = 0^+$, production $m' \approx \nu - \mu m$ (or $m' \approx \nu$ while accumulating) at the late fully-ON rate ν , Tat switch-on folded into a slightly longer eclipse plus an abrupt start of ν , and the export law left open — the linear-vs-exponential

choice for accumulation does not force it. In ranking: birth–death / absorbing BDC fits poorly as a literal HIV mechanism; immigration plus export or dump is a much better caricature; eclipse then constant production rate is the best simple default; Tat ON/OFF plus immigration the best simple mechanistic default.

16.5 Suggested role of this material

The core lytic results stand as the chapter’s contribution; this section is their honest boundary. In a companion paper the natural order is: the BDC and renewal results at the lytic end of the spectrum; then “HIV is not BDC”, with Hataye’s Model 3 as the empirical intermediate and the stage equations above; then numerical extraction of $g_{\text{HIV}}(a)$ from the stage model (or from daily release curves), overlaid against constant- p BMVR exactly as in §12.4; and finally establishment by independent branching vs the Allee form, with the critical release scale taken from experiment.

17 Discussion

The chapter set out to complete the single-cell theory of the birth–death–catastrophe process and to scale it up into a population model forced by the intracellular dynamics. Both are done: the single-cell description is now complete in distribution — geometric states at every time, a geometric quasi-stationary limit, and a burst-size law identical to it — and the population model exists, verified, and carries exact consequences for fitting, for thresholds, and for the evolutionary question of bursting versus budding. What remains is to say what the results imply for measurement, how they connect to the chapters that follow, and what is still open.

17.1 Single-cell assay predictions for lytic systems

For a genuinely lytic system — *Y. pestis*, *F. tularensis*, lytic phage — a limiting-dilution or Hataye-type assay (one infected cell per well, counts followed in time) measures the single-cell process directly, with no population model interposed. The theory makes five predictions, all of them parameter-free given (β, μ, δ) :

1. a fraction b of wells show no release ever — the mass of internal extinction;
2. the cumulative release in a well is a step function: zero, then a single terminal jump;
3. the jump times follow the density $\delta J(t)/(1 - b)$;
4. the jump sizes follow the geometric law with ratio $1/a$ — the quasi-stationary distribution, by Corollary 7.2;
5. the between-well variance of total release is $\text{Var}(W_\infty)$ of (13), a quantity no deterministic constant-rate model can produce at all.

The honest caveat of §16 applies: these predictions are too strong for HIV as literally stated — intermittent budding is not a single terminal burst — and they should be used for lytic hosts, or as the bursting endpoint of the spectrum of §15.

17.2 Consequences for fitting practice

Three consequences for the way within-host models are fitted to data. First, p is not a constant of the pathogen: the effective release rate $p_{\text{eff}}(r)$ falls monotonically with the epidemic growth rate, from $V_{\infty}/\mathbb{E}(T_{\text{prod}})$ in the stationary phase towards δ in the acute phase, and fits to acute and set-point data should therefore yield systematically different estimates, in a direction and by an amount the theory specifies (§13). Second, the eclipse-like delay usually inserted into within-host models by hand emerges from the intracellular dynamics with no extra parameter; its duration is the mean productive lifetime’s shadow, $\beta^{-1} \log \langle X \rangle_{\text{QS}}$. Third, the identifiability negative result of §13.3 says that no fit of $(p, d_{\mathcal{I}})$ — from any data, at any growth rate — can recover the three intracellular rates; a third observable is required, and the theory names the three cheapest ones (burst-size dispersion, the no-burst fraction b , the shape of the release-time density). An estimate of p alone tells one almost nothing about the cell; an estimate of p plus any one of the three tells one something testable.

17.3 Connections forward

The two-type process of Chapter 5 supplies the most immediate continuation. In the GATE regime $\delta_1 = 0 < \delta_2$ — no burst before conversion — the no-catastrophe probability has $S'(0) = 0$: a zero initial hazard, then rising. That is precisely the eclipse-phase signature which within-host models insert by hand, emerging mechanistically, with the eclipse duration set by the conversion rate ν . The renewal construction of this chapter is ready to receive the two-type kernels; what blocks the way is the two-type calculation itself (below). The budding–bursting comparison of §14 finds its evolutionary home in Chapter 6, where the spectrum of §15 — partial release, reset catastrophe, self-excitation — provides the intermediate phenotypes a selection argument needs; and the nonlinear mechanisms Chapter 7 discusses in the Komarova vein (immune saturation, local depletion, the “all run out at once” intuition) are exactly the effects the branching-structure comparison of §14 deliberately excludes, and should be named wherever the flooding advantage is invoked.

17.4 Open problems

1. **A conceptual proof that burst size equals QSD.** The algebra of §7.3 is complete; a structural argument — presumably size-biasing composed with the burst intensity, or a Yaglom-type limit [5] — would turn a coincidence into a theorem one can quote.
2. **Two-type killed second moments and the two-type D .** The release kernel of a two-type renewal model needs $\mathbb{E}(X_a^2 \mathbf{1}\{\tau_c > a\})$, $\mathbb{E}(X_a Y_a \mathbf{1}\{\tau_c > a\})$, $\mathbb{E}(Y_a^2 \mathbf{1}\{\tau_c > a\})$, and the analogue of $D(t)$ (extinction without catastrophe). The moment hierarchy does not close upward, exactly as in one type; the one-type escape was to differentiate the Riccati equation, and the two-type route should be to differentiate the backward system of Chapter 5. This is the main technical obstacle between this chapter and an eclipse-aware renewal BMVR, and it should not be assumed easy.
3. **The flooding boundary against real parameters.** Map $\delta(\beta + \mu) > \beta\mu$ for *Y. pestis* in the macrophage, and compute the *size* of the advantage from (98): does the pathogen sit on the advantageous side, and by how much?
4. **Population-level variance.** Propagate $\text{Var}(W)$ of (13) through the branching structure to a whole-host prediction: for single-cell assays the first moments suffice, but for whole-host data the variance is the main stochastic signal the BMVR cannot reproduce.

5. **Partial release and the flooding boundary in q .** Where on the boolean continuum of §15.5 does the flooding advantage survive? Does the closed-form structure persist at intermediate q ?
6. **Logistic intracellular growth.** A referee will object that linear birth lets the conditional load grow without bound. The defence is that the burst hazard is proportional to load and so regulates it — the QSD result is precisely the statement that the conditional load stabilises at $a/(a-1)$ — but the sensitivity of $g(a)$ and $\hat{I}(a)$ to a logistic birth rate remains to be checked numerically.
7. **Literature positioning.** Verify whether Carruthers et al. [12] already couple an intracellular BDC to a between-cell model — the novelty claim of §12 depends on the answer — and position the renewal embedding against the Pearson–Krapivsky–Perelson [13], Nelson–Gilchrist–Perelson, Gilchrist–Coombs [14], and Heffernan–Wahl lines of work, with the BMVR’s attribution settled between McLean, Nowak–Bangham, and Perelson et al.

A Master formula tables

Single cell

Throughout: $w = e^{\theta t}$, $\theta = \beta(a-b)$, $A = a-1$, $B = 1-b$, $L = a(1-b)$, $\kappa = 1 + \delta/(2\beta)$.

Quantity	Closed form
Roots a, b	$\eta \pm \sqrt{\eta^2 - \mu/\beta}$, $\eta = (\beta + \mu + \delta)/(2\beta)$
A, B, θ	$a-1, 1-b, \beta(a-b)$; $AB = \delta/\beta$, $A+B = a-b$
$I(t)$	$(aB + bAw)/(B + Aw)$
$D(t)$	$ab(w-1)/(aw-b)$
$\hat{I}(t)$	$(a-b)^2 w / [(B + Aw)(aw-b)]$ (corr.)
$\hat{I}_k(t)$	$I(t)^k - D(t)^k$ (corr.)
$J(t)$	$(a-b)^2 w / (B + Aw)^2 = -\frac{\beta}{\delta}(I-a)(I-b)$
$K(t)$	$[1 + 2\beta(1-I)/\delta]J = \frac{2\beta}{\delta}(\kappa - I)J$
$V(t)$	$(1-I)[1 + \beta(1-I)/\delta]$
V_∞	$a(1-b)/(a-1) = aB/A$
$\mathbb{E}(\mathcal{K} \mid \text{burst})$	$a/(a-1)$
$\mathbb{E}(\mathbf{W}_t^2)$	$\frac{2(\beta-\mu)}{\delta}V - \frac{\beta+\mu}{\delta}I - K + \frac{\beta+\mu}{\delta} + 1$ (corr.)
$\varphi(t)$ (burst time)	$\delta J(t)$, mass $1-b$
$\mathbb{P}\{\mathcal{K} = k\}$	$(\delta/\beta) a^{-k}$, $k \geq 1$
QSD(n)	$(a-1) a^{-n}$ (new)
$\mathbb{E}(T_{\text{prod}})$	$\beta^{-1} \log(a/(a-1)) = \beta^{-1} \log \langle X \rangle_{\text{QS}}$ (new)
$g_k(a)$ (MOI)	$\delta[kKI^{k-1} + k(k-1)J^2I^{k-2}]$ (corr.)
$V_\infty^{(k)}$, $\mu = 0$	$k + \beta/\delta$ (new)
r_k law ($\mu = 0$, chain)	$\binom{n+k-2}{k-1} s^k \rho^{n-1}$, $s = \delta/(\beta + \delta)$
$\tilde{T}(k, u)$ (interval)	$\frac{sk}{k+u'} {}_2F_1(k+1, 1; k+1+u'; \rho)$, $u' = u/(\beta + \delta)$

Population

Quantity	Formula
Renewal system	$\mathcal{I} = \mathcal{I}_0 \hat{I} + i * \hat{I}, \quad \mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}, \quad i = \gamma T\mathcal{V}$
Release kernel	$g(a) = \delta K(a) = V'(a)$
$p_{\text{eff}}(r)$	$\delta \tilde{K}(r) / \tilde{I}(r)$
$d_{\mathcal{I}, \text{eff}}(r)$	$1 / \tilde{I}(r) - r$
Limits $r = 0$	$p_{\text{eff}} = \beta V_{\infty} / \log(a/(a-1)), \quad d_{\text{eff}} = \beta / \log(a/(a-1))$
Limits $r \rightarrow \infty$	$p_{\text{eff}} \rightarrow \delta, \quad d_{\text{eff}} \rightarrow \mu + \delta$
Characteristic eq.	$r + c = \gamma T \delta \tilde{K}(r)$
R_0^{ODE}	$\gamma T V_{\infty} / c$
Branching mean	$m = q V_{\infty}, \quad q = \gamma T / (\gamma T + c)$
$G_{\text{off}}(z)$	$b + (\delta/\beta) y / (a - y), \quad y = 1 - q + qz$
$z_{\text{ext}}^{\text{burst}}$	$[a(1 - q + qb) - 1 + q] / q$
Flooding criterion	$1/\delta < 1/\beta + 1/\mu \quad (L > 1)$
$L = 1$	offspring law exactly geometric
Δz_{ext}	$(L - 1)(1/m - 1)$
$\text{Var}_{\text{bud}} - \text{Var}_{\text{burst}}$	$2q^2 V_{\infty} (L - 1) / (a - 1)$
Growth trade-off	$r_{\text{bud}} > r_{\text{burst}}$ at matched $R_0, \mathbb{E}(T_{\text{prod}})$

Quick-reference specialisations

$\mu = 0$. $b = 0, a = 1 + \delta/\beta, D \equiv 0, \hat{I} = I, V_{\infty} = 1 + \beta/\delta = \langle X \rangle_{\text{QS}}, \mathbb{E}(T_{\text{prod}}) = \beta^{-1} \log(1 + \beta/\delta), V_{\infty}^{(k)} = k + \beta/\delta, \mathbb{P}\{\mathcal{K} = k\} = s\rho^{k-1}$ with $s = \delta/(\beta + \delta), \rho = 1 - s$; flooding favours bursting for establishment at every $\delta > 0$.

The boundary $L = 1$. $\delta = \beta\mu/(\beta + \mu), b = \delta/\beta$; the offspring law is geometric with mean m ; $z_{\text{ext}}^{\text{burst}} = z_{\text{ext}}^{\text{bud}}$ for every q ; offspring variances equal.

Young-cell / fast-growth limit. $p_{\text{eff}} \rightarrow \delta, d_{\text{eff}} \rightarrow \mu + \delta$: a newly infected cell holds one particle, bursts at rate δ , releases one.

Set-point / $r = 0$ matching. $p_{\text{eff}} = V_{\infty}/\mathbb{E}(T_{\text{prod}}), d_{\text{eff}} = 1/\mathbb{E}(T_{\text{prod}})$: lifetime output over lifetime; the fairest constant-parameter comparison of §12.4.

B Verification records

Every result in this chapter was checked analytically where possible and numerically where not. This appendix records the two numerical suites, their tallies as of the last run (2026-08-07), and how to reproduce them.

B.1 Renewal BMVR suite (Result 12.1)

The suite `verify_result_20_1.py` tests the renewal system (81)–(82) at four levels: the integrity of the kernels; the reduction to single-cell formulae without reinfection; the structural limits that can be checked in closed form; and an independent stochastic simulation of a finite cohort. Parameter sets: $(\beta, \mu, \delta) \in \{(1, 0, 0.1), (1, 0.2, 0.05), (1, 0, 0.5), (1, 0.9, 0.1), (1, 0.5, 1/3)\}$.

ID	Name	What is checked	Result
A	Kernel self-consistency	$V' = g, \int g = V_\infty, \int \hat{I} = \mathbb{E}(T_{\text{prod}}), I' = -\delta J$	5/5
B	$\gamma T = 0$ cohort	$\mathcal{I} = \mathcal{I}_0 \hat{I}; \mathcal{V}' = \mathcal{I}_0 g - c\mathcal{V}$ vs exact convolution; $c = 0 \Rightarrow \mathcal{V} = \mathcal{I}_0 V(t)$	15/15
C	V -equation residual	$\mathcal{V}' - \mathcal{I}_0 g - \gamma T(\mathcal{V} * g) + c\mathcal{V} \approx 0$ on supercritical / subcritical / virion-seed solutions	5/5
D	Exponential kernels $\equiv$ classical	$S = e^{-da}, g = pe^{-da} \Rightarrow$ renewal = BMVR ODE (rel. err. $< 5 \times 10^{-5}$)	3/3
E	Growth = characteristic root	late log $\mathcal{V}$ slope equals r_* solving $r+c = \gamma T \delta \tilde{K}(r)$ (rel. err. $\sim 10^{-6}$ – 10^{-7})	5/5
F	R_0 threshold	$R_0 \in \{0.5, \dots, 3\}$: decay iff $R_0 < 1$, growth iff $R_0 > 1$	8/8
G	$p_{\text{eff}}(r_*)$ match	classical ODE with $(p, d) = (p_{\text{eff}}(r_*), d_{\text{eff}}(r_*))$ shares r_* with the renewal system	4/4
H	Gillespie cohort	independent BDC cells + clearance; ensemble mean vs renewal (rel. $L^2 \sim 1$ –2%)	2/2
I	Virion-only seed	early $\mathcal{V} \approx V_0 e^{-ct}; \mathcal{I}(0) = 0$ then rises	1/1
J	Lifetime output	$\gamma T = c = 0: \mathcal{V}(\infty) = V_\infty$ (rel. err. $< 10^{-6}$)	5/5
K	Well-posedness	non-negative, finite trajectories	1/1
Total			54/54

(Earlier summaries of this suite quote 53 checks; the present version adds one well-posedness check.) One honest gap: a full stochastic multi-generation Gillespie check with reinfection ($\gamma T > 0$) is not yet coded. The reinfection case is already constrained by tests C, E, F, G through the characteristic equation, which any correct linear theory must satisfy, and agreement is expected by linearity.

Reproduction.

```
cd "4 BDC additional and BMVR/new_notes3"
python3 verify_result_20_1.py
# writes verify_result_20_1_report.txt, metrics, verify_figures/*.pdf
```

B.2 Chained immediate-transfer suite (§10)

The suite `verify_chained_transfer.py` samples the chained process exactly (the event-type decomposition of §10.2 makes simulation direct) and checks every quantitative claim of §10 across three parameter sets, $(\beta, \delta) \in \{(1, 1), (1, 0.1), (2, 0.5)\}$.

ID	Name	What is checked	Result
A	Coefficient identity	$\binom{n+k-2}{k-1}$ obeys the recurrence (exact integers)	1/1
B	Rupture-size laws	$\mathbb{P}\{r_k = n\}$ vs 10^6 chains, $k = 1..4$, SE-unit comparison	3/3
C	Moments	$\mathbb{E}(r_k) = 1 + k\beta/\delta$, $\text{Var}(r_k) = k\rho/s^2$	3/3
D	PGF	$\mathbb{E}(z^{r_k}) = z(s/(1 - \rho z))^k$ (SE units)	3/3
E	First rupture time	histogram vs $\delta J(t)$	3/3
F	Interval transform	fixed founders $k \in \{1, 2, 3, 5\}$: empirical Laplace transform vs the series ${}_2F_1$ form	3/3
F2	Interval mixture	chain intervals T_2, T_3 (random founders) vs pmf-weighted $\tilde{T}$	3/3
G	Second rupture time	empirical $\mathbb{E}(e^{-ut_2})$ vs $\tilde{t}_2(u)$	3/3
H/H2	t_2 density and mean	exact hypoexponential-mixture density vs histogram (rel. $L^2 \approx 3\%$); exact $\mathbb{E}(t_2)$	3/3
I	Interval means	strictly decreasing; simulation vs the exact series	3/3
Total			28/28

The ${}_2F_1$ cross-check in test F activates only when scipy is available; the series itself (which the simulation is compared against) needs only numpy.

Reproduction.

```
cd "4 BDC additional and BMVR/N=2 Immediate Transfer content"
python3 verify_chained_transfer.py
# writes verify_chained_transfer_report.txt, metrics, figures
```

C Technical derivations

C.1 Hypergeometric Laplace transforms of the kernels

Write $s := r/\theta$ and recall $\theta = \beta(a - b)$. Both transforms reduce, under $v = e^{-\theta t}$ (so $dt = -dv/(\theta v)$, $v : 1 \rightarrow 0$), to moments of the form $\int_0^1 v^{s-1}(1 - v)^n(A + Bv)^{-m}dv$, which are hypergeometric. For the survival kernel, from (7),

$$\hat{I}(t) = \frac{(a - b)^2 w}{(B + Aw)(aw - b)}, \quad w = e^{\theta t},$$

and in $v = 1/w$,

$$\tilde{I}(r) = \frac{a - b}{\theta} \int_0^1 v^s \left[\frac{B}{A + Bv} + \frac{b}{a - bv} \right] dv,$$

using the partial-fraction decomposition $(a - b)/[(A + Bv)(a - bv)] = B/(A + Bv) + b/(a - bv)$ (both sides multiplied out give $(a - b)^2$ in the numerator after using $Ab + aB = a - b$). Each term is a standard hypergeometric integral, $\int_0^1 v^s/(1 - zv)dv = (s + 1)^{-1} {}_2F_1(1, s + 1; s + 2; z)$, giving

$$\tilde{I}(r) = \frac{1}{\beta(s + 1)} \left[\Psi(A, B, s) + \frac{b}{a} {}_2F_1\left(1, s + 1; s + 2; \frac{b}{a}\right) \right], \quad (133)$$

where, so that the hypergeometric series always converges,

$$\Psi(A, B, s) = \begin{cases} {}_2F_1(1, s+1; s+2; -\frac{A}{B}), & B \geq A, \\ \frac{B}{A} {}_2F_1(1, s+1; s+2; -\frac{B}{A}), & A > B, \end{cases} \quad (134)$$

the two branches being equal by the identity $(B/A){}_2F_1(1, s+1; s+2; -B/A) = {}_2F_1(1, s+1; s+2; -A/B)$ — both sides equal $(s+1)B \int_0^1 v^s / (A+Bv) dv$. (The single-argument form $(B/A){}_2F_1(1, s+1; s+2; -B/A)$, valid by analytic continuation for all $A, B > 0$, has a convergent defining series only when $B \leq A$, i.e. $\delta \geq \beta - \mu$.) For the release kernel, $g = \delta K$ with $K = [1 + 2\beta(1-I)/\delta]J$; the same substitution and termwise integration give

$$\delta \widetilde{K}(r) = \frac{\delta}{\beta} \left[\frac{{}_2F_1(1, s; s+2; z)}{A(s+1)} + \frac{2 {}_2F_1(2, s; s+3; z)}{A^2(s+1)(s+2)} \right], \quad z = -\frac{B}{A}, \quad (135)$$

again with the understanding that for $B > A$ the defining series of the right-hand side must be evaluated through its analytic continuation (or through the integral representation from which it came). Both formulae reduce correctly at $s = 0$: with ${}_2F_1(1, 1; 2; z) = -\log(1-z)/z$, (133) gives $\widetilde{I}(0) = \beta^{-1} \log(a/(a-1)) = \mathbb{E}(T_{\text{prod}})$, and (135) gives $\delta \widetilde{K}(0) = (\delta/\beta)(1/A + 1/A^2) = aB/A = V_\infty$, using $AB = \delta/\beta$. Both reductions were verified numerically against direct quadrature, as was (135) for general r (relative error $\sim 10^{-8}$ in the convergent regime).

C.2 Lifetime yield from k founders

We derive (54). Start from $V_\infty^{(k)} = \int_0^\infty g_k(t) dt$ with g_k from (52), and change variables from t to $x = I(t)$: as t runs $0 \rightarrow \infty$, I runs $1 \rightarrow b$, and since $dI/dt = -\delta J$, $dt = -dx/(\delta J)$. The first part of g_k gives

$$\int_0^\infty \delta k K I^{k-1} dt = \int_b^1 k \frac{K}{J} x^{k-1} dx = \int_b^1 k \left(1 + \frac{2\beta}{\delta} - \frac{2\beta}{\delta} x \right) x^{k-1} dx,$$

using $K/J = 1 + 2\beta(1-I)/\delta$. Integrating,

$$= \left(1 + \frac{2\beta}{\delta} \right) (1 - b^k) - \frac{2\beta k}{\delta} \frac{1 - b^{k+1}}{k+1} = (1 - b^k) + \frac{2\beta}{\delta} \left[\frac{1}{k+1} - b^k + \frac{k b^{k+1}}{k+1} \right],$$

where the second line collects terms over the common denominator $k+1$. The second part of g_k gives, with $J = -\frac{\beta}{\delta}(I-a)(I-b)$,

$$\int_0^\infty \delta k(k-1) J^2 I^{k-2} dt = \int_b^1 k(k-1) J x^{k-2} dx = k(k-1) \frac{\beta}{\delta} \int_1^b (x-a)(x-b) x^{k-2} dx,$$

on reversing the limits. Adding the two parts yields (54). The formula also holds at $k = 1$ (the integral term vanishes), reducing to $V_\infty = (1-b)[1 + \beta(1-b)/\delta]$ of (12). When $\mu = 0$, $b = 0$ and the integral simplifies to $k(k-1) \frac{\beta}{\delta} \int_1^0 (x-a) x^{k-1} dx = k(k-1) \frac{\beta}{\delta} \left[\frac{a-1}{k} - \frac{1}{k+1} \right]$, which combines with the first part to give $k + \beta/\delta$ — the elementary result of §9, recovered analytically.

C.3 PGF coefficient extraction for general initial conditions

The general- μ route for the chained model of §10 runs through the k -founder probability generating function. From §4.2 (catastrophe convention),

$$G_k(z, t) = \left[\frac{ab\gamma(t) + v(t)z}{1 - \gamma(t)z} \right]^k, \quad \gamma(t) = \frac{1 - e^{\theta t}}{b - ae^{\theta t}}, \quad v(t) = \frac{be^{\theta t} - a}{b - ae^{\theta t}}.$$

Writing $\Theta_m = \binom{k}{m}(ab)^{k-m}\gamma(t)^{k-m}v(t)^m$ and $\Omega_n = \binom{k-1+n}{n}\gamma(t)^n$, the binomial expansion of the numerator and the negative-binomial series of $(1-\gamma z)^{-k}$ give $G_k(z, t) = (\sum_{m=0}^k \Theta_m z^m)(\sum_{n \geq 0} \Omega_n z^n)$, so the state-probability coefficients are, with H the power of z ,

$$[z^H]G_k = \sum_{i=0}^{\min(H,k)} \Theta_{k-i} \Omega_{i+\max(0,H-k)}.$$

The chained burst-size laws then require integrals of the form $\int_0^\infty \gamma(t)^n v(t)^m dt$ for arbitrary $n, m \in \mathbb{N}$. Binomial-expanding the two numerators,

$$(1 - e^{\theta t})^n (b e^{\theta t} - a)^m = \sum_{i=0}^{n+m} K_i (e^{\theta t})^i$$

for explicit constants K_i , the problem reduces to the elementary building blocks

$$\int \frac{(e^{\theta t})^i}{(b - a e^{\theta t})^j} dt = -\frac{i (e^{\theta t})^i (b - a e^{\theta t})^{1-j}}{b \theta} {}_2F_1\left(1, (1+i) - j; 1+i; \frac{a e^{\theta t}}{b}\right),$$

and the burst-size law for the k th rupture follows by summing over the initial-condition distribution. For $\mu = 0$ this entire machinery collapses to the geometric decomposition of §10.2; for $\mu > 0$ it remains the available route, and the joint laws of (t_k, r_k) are the open problem stated in §10.7.

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